Q3 Series Q

Hands-on II: multi-sample analysis integration, cell identity, DE and beyond

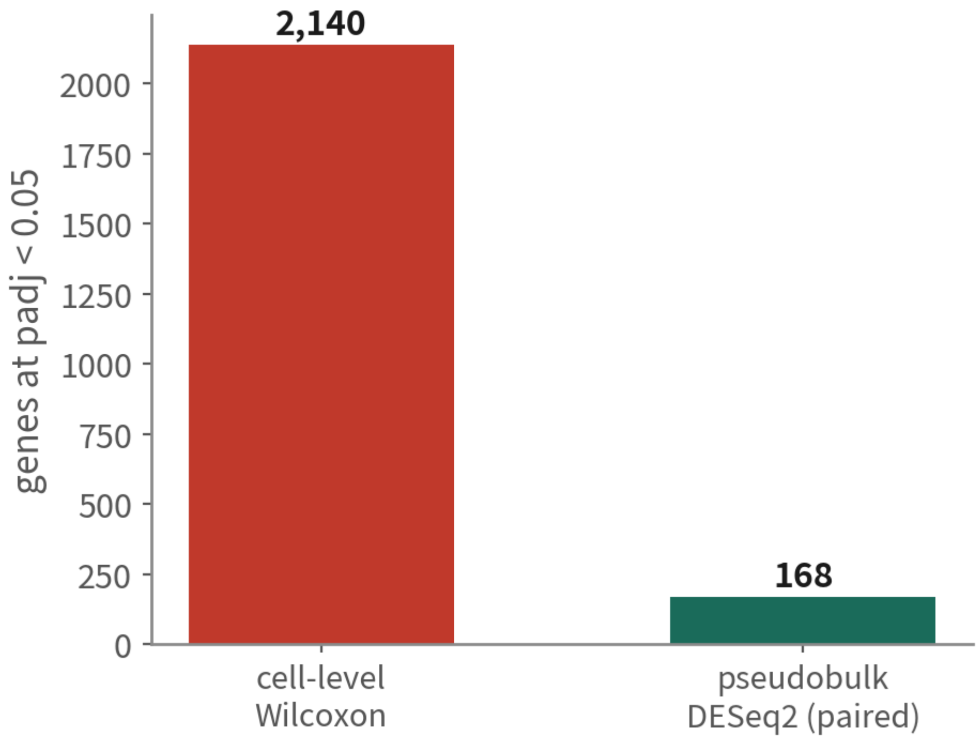
Four GBM patients, core vs infiltrating periphery. Fill in Q2's 'pending CNV', make DE patient-level, then pick the next road.

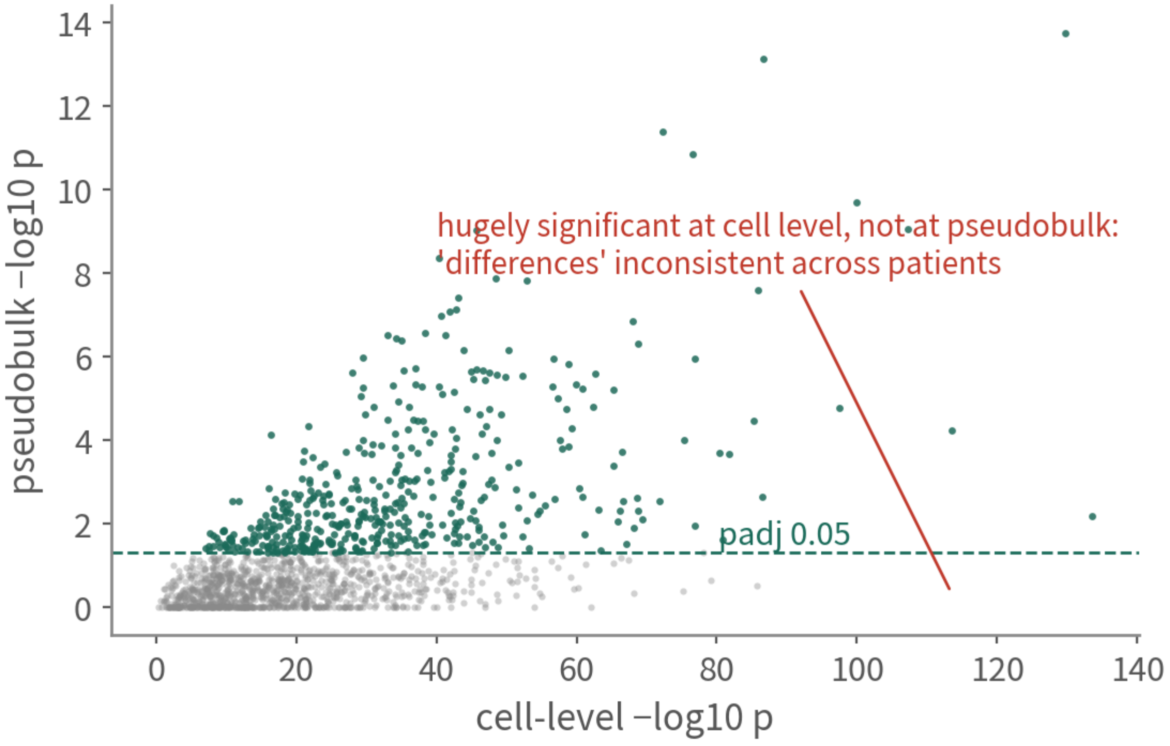
About 70 min | Prerequisites: Q2 | Scripts: R/04–10 | Deeper: B11–B17

Same question, two answers

Same question (core vs periphery malignant cells), two answers

schematic / simulated





Which genes differ between core and periphery malignant cells? Cell-level says two thousand, pseudobulk says a hundred-odd. In 65 minutes you will know which to trust

WHERE THIS EPISODE STARTS

Multi-patient tumour data comes down to three decisions: how far to integrate, who is malignant, and what the unit of DE is.

Get all three right and the result survives review; get one wrong and no figure will save it.

What you will do

1. Combine four patients with Seurat's default integration (Harmony as the comparison), and use positive / negative controls to check under- and over-correction
2. Call malignant cells with inferCNV using immune and oligodendrocyte references, triangulate, write to metadata, check against the authors' labels
3. Run composition analysis, pseudobulk + DESeq2 (paired), GSEA, and use one table to choose the next downstream road

01

Four patients, eight samples

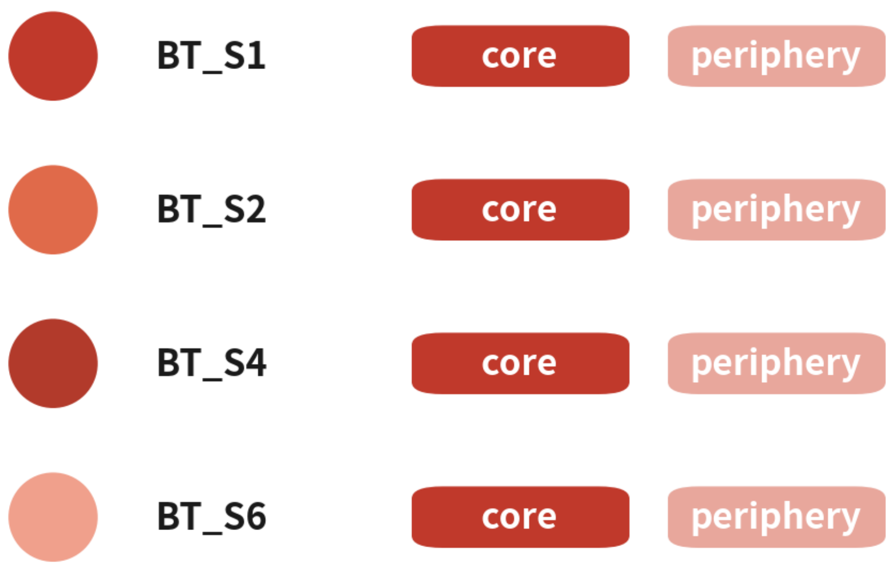
Script 04: load Smart-seq2 counts and GEO metadata

Practice data: Darmanis et al. 2017

Practice data: Darmanis et al. 2017 (GSE84465)

tissue	4 GBM patients, each sampled at tumour core and infiltrating periphery
cells	3,589 (author QC passed)
method	Smart-seq2 full-length (no UMI; read counts)
labels	7 types: neoplastic, OPC, astrocyte, oligodendrocyte, neuron, vascular, immune
why	multi-patient+a condition (core vs periphery)+malignant labels to check against

Design: 4 × 2 paired



Two sites from the same patient can be compared as pairs: n=4 pairs

Three things Q2 lacked: multiple patients, a condition (core vs periphery), and author malignant labels to check against

Smart-seq2 vs 10x: what differs, what does not

	10x 3' (Q2's GBM 5k)	Smart-seq2 (Q3's GSE84465)
unit	UMI molecules	read counts (PCR duplicates not removed)
total per cell	5k–30k	0.5–several million reads
gene coverage	3' end, 1–3k genes	full length, 4–8k genes
cells / cost	many, cheap	few, expensive, information-rich
QC differences	percent.mt works; catch doublets	percent.mt still works; one cell per well, few doublets
Seurat pipeline	identical	identical (NormalizeData applies)
pseudobulk	sum UMI counts	sum read counts (still integers; DESeq2 works)

The Seurat pipeline is identical; the differences are the counting unit, information per cell, and that doublets are barely a concern

Load counts and metadata

```
# GEO 補充檔（約 20 MB）：基因 × 細胞的 read counts
cnt <- read.csv("data/GSE84465_GBM_All_data.csv.gz",
                row.names = 1, check.names = FALSE)
bad <- c("no_feature", "ambiguous", "alignment_not_unique",
         "too_low_aQual", "not_aligned")        # HTSeq 統計列
cnt <- cnt[!rownames(cnt) %in% bad, ]

library(GEOquery)                  # metadata 在 series matrix
gse  <- getGEO("GSE84465", GSEMatrix = TRUE)[[1]]
meta <- pData(gse)
# characteristics 欄位順序不保證，先 head(meta) 確認
meta$patient <- sub("patient id: ", "", meta$characteristics_ch1.3)
meta$tissue  <- sub("tissue: ", "", meta$characteristics_ch1.2)
meta$celltype_author <- sub("cell type: ", "", meta$characteristics_ch1.4)
```

Column order in characteristics is not guaranteed; head(meta) first. Script 04 finds the columns with grep | Script 04 §1

Build the object, quick QC

```
# 細胞名稱 = plate.well（例：1001000173.G8），與 counts 的欄名對齊
meta$cell <- paste(sub("plate id: ", "", meta$characteristics_ch1.1),
                   sub("well: ", "", meta$characteristics_ch1.2), sep = ".")
meta <- meta[match(colnames(cnt), meta$cell), ]

gbm4 <- CreateSeuratObject(counts = as.matrix(cnt), meta.data = meta,
                           min.cells = 3, min.features = 500)
gbm4[["percent.mt"]] <- PercentageFeatureSet(gbm4, pattern = "^MT-")
VlnPlot(gbm4, c("nFeature_RNA", "percent.mt"), group.by = "patient")
table(gbm4$patient, gbm4$tissue)
```

```
       Periphery Tumor
BT_S1        ...   ...     ← a few hundred per cell; all eight samples present
```

Smart-seq2 nFeature runs 4–8k, so the floor is 500 not 200; the authors already QC'd, this is confirmation | Script 04 §2

02

Integration: how much correction is enough

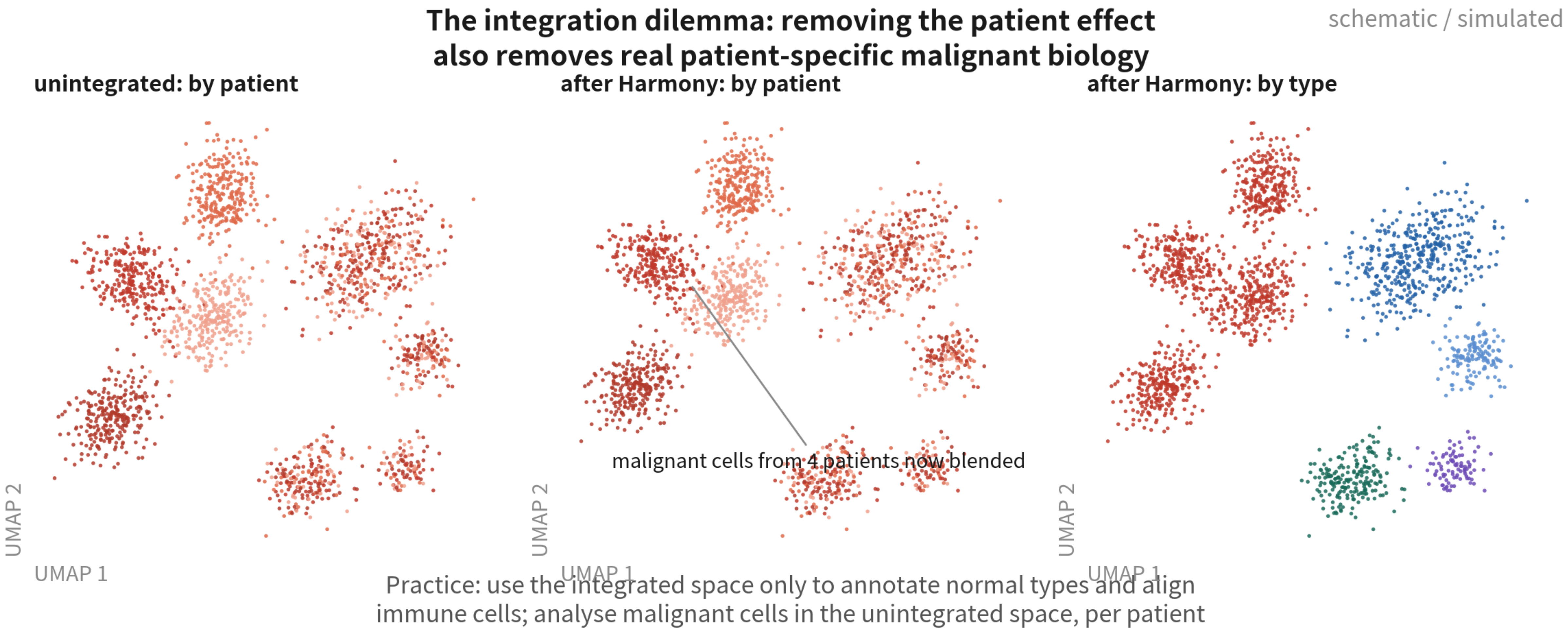
Integration is a dilemma in tumour data; look at the unintegrated plot first

Run the pipeline once without integration and keep the baseline

```r
gbm4 <- NormalizeData(gbm4) |> FindVariableFeatures() |> ScaleData() |> RunPCA()
gbm4 <- FindNeighbors(gbm4, dims = 1:25) |> FindClusters(resolution = 0.5)
gbm4 <- RunUMAP(gbm4, dims = 1:25, reduction.name = "umap.raw")
DimPlot(gbm4, reduction = "umap.raw",
        group.by = c("patient", "celltype_author"))
saveRDS(gbm4, "output/gbm4_unintegrated.rds")      # 之後惡性分析回來用這份
```

Without a baseline you cannot tell what integration changed. Author labels serve as the known answer here | Script 04 §3 | Deeper: B11

The unintegrated plot already tells you something



Immune and oligodendrocyte cells mix across patients; malignant cells form patient islands. Integration blends the latter too — and that is real biology

The integration map: how hard it corrects, what it needs, the risk for tumours

There is no best method, only a correction strength that matches the question

Method	Idea	Strength	Best for	Risk in tumour data
Seurat CCA / RPCA (default here)	Anchor pairs of cells across samples; built into Seurat v5, one IntegrateLayers call	RPCA weak CCA strong	Samples with high type overlap	CCA over-corrects most; patient-private types get merged into others
Harmony (comparison)	Iteratively pulls batch centroids together in PC space	medium	Many patients / batches fast, light	High theta forces malignant cells together
scVI / scANVI	Variational autoencoder with batch as covariate	tunable	Large atlases, cross-technology (10x + Smart-seq2)	Needs GPU and Python latent space is hard to interpret
No integration, analyse apart	Integrated space for normal cells; malignant cells per patient	—	The mainstream tumour approach	Comparing malignant states across patients needs scores (Neftel), not clusters

Integration: Seurat default first, Harmony to compare

```
gbm4[["RNA"]] <- split(gbm4[["RNA"]], f = gbm4$patient)   # 每位病人一個 layer
gbm4 <- NormalizeData(gbm4) |> FindVariableFeatures() |> ScaleData() |> RunPCA()
# (a) Seurat 預設：CCA 錨點（大資料可換 RPCAIntegration）
gbm4 <- IntegrateLayers(gbm4, method = CCAIntegration,
                        orig.reduction = "pca", new.reduction = "integrated.cca")
# (b) 比較：Harmony（Seurat 整合不理想時的替代方案）
gbm4 <- IntegrateLayers(gbm4, method = HarmonyIntegration,
                        orig.reduction = "pca", new.reduction = "harmony")
gbm4 <- JoinLayers(gbm4)
for (red in c("integrated.cca", "harmony")) {   # 同 dims / resolution 才能比
  tag <- sub("integrated.", "", red, fixed = TRUE)
  gbm4 <- FindNeighbors(gbm4, reduction = red, dims = 1:25,
                        graph.name = paste0(tag, "_snn")) |>
          FindClusters(graph.name = paste0(tag, "_snn"), resolution = 0.5,
                       cluster.name = paste0(tag, "_clusters")) |>
          RunUMAP(reduction = red, dims = 1:25,
                  reduction.name = paste0("umap.", tag))
}
DimPlot(gbm4, reduction = "umap.cca", group.by = "patient") |
DimPlot(gbm4, reduction = "umap.harmony", group.by = "patient")
```

Integration only adds reductions; no expression value changes; DE always returns to RNA counts. Judge Seurat's result first; switch to Harmony only when the positive control fails (immune cells still split by patient) | Script 04 §4 | Deeper: B11

Positive and negative controls for integration: what should blend, what should stay apart

Positive and negative controls for integration: what should blend, what should stay apart

	expected (pass)	warning sign (fail)
positive control: immune	✓ four patients blend →integration worked	⚠ islands by patient→ batch not corrected
positive control: oligo, vascular	✓ mix across patients	⚠ islands by patient
negative control: malignant	✓ islands by patient (correct!)	⚠ blended→over-corrected; patient-specific CNV biology erased

Positive control: immune and oligodendrocyte cells should blend across patients (no blend = under-corrected); negative control: malignant cells should keep patient differences (blended = over-corrected). Apply the same controls to both integrations

Turn 'did it blend' into numbers: per-cluster patient mix, LISI

```r
# 1. 每個整合群裡各病人佔多少：正常細胞的群應該四位都有
round(prop.table(table(gbm4$cca_clusters, gbm4$patient), 1), 2)

# 2. LISI：一顆細胞的鄰居裡「有效病人數」（1 = 全是同一位，4 = 四位均勻）
library(lisi)
for (red in c("pca", "integrated.cca", "harmony")) {
  emb <- Embeddings(gbm4, red)[, 1:25]
  gbm4[[paste0("lisi_", sub("integrated.", "", red, fixed = TRUE))]] <-
    compute_lisi(emb, gbm4@meta.data, "patient")$patient
}
gbm4@meta.data |> group_by(celltype_author) |>
  summarise(across(starts_with("lisi_"), ~ median(.x)))
```

```
celltype_author  lisi_pca  lisi_cca  lisi_harmony
Immune cell          ~1.x      ~3.x      ~3.x     ← positive control: well mixed
Oligodendrocyte      ~1.x      ~3.x      ~3.x
Neoplastic           ~1.0      ~1.x      ~2.x     ← negative control: near 1 = patient signal kept
```

Immune LISI near the patient count = integration worked; malignant LISI near 1 = patient differences preserved. Report both and reviewers stop guessing | Script 04 §5

THE INTEGRATION RULE FOR TUMOUR DATA

Use the integrated space only to annotate normal cells and align immune cells;

analyse malignant cells in the unintegrated space, per patient.

Not a compromise — the two populations have different statistical structure. Most tumour papers (Tirosh, Neftel, Puram) do exactly this.

The other road: map cells onto a reference atlas

With a good reference you need not annotate from scratch — normal cells especially

Azimuth

Project your cells onto an annotated reference (PBMC, human brain, lung…) and inherit labels and UMAP coordinates. The easiest route for immune subsets

scArches

Fine-tune the reference model (scVI / scANVI) on your data without re-training the atlas; works across technologies. Python ecosystem

Malignant cells cannot be mapped

No atlas contains this patient's tumour. Mapped malignant cells only get 'the nearest normal type' — like SingleR, a reminder, not an answer

03

Cell identity: calling malignancy with inferCNV

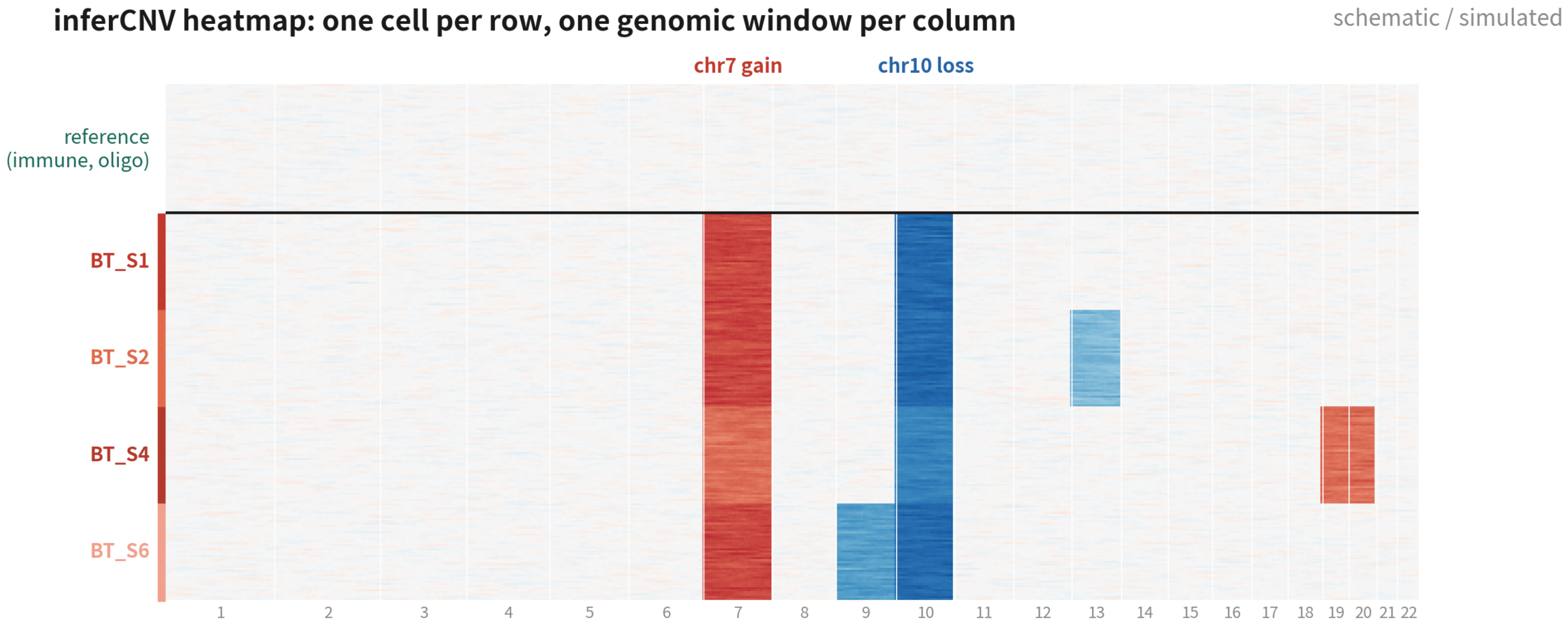
Script 05: fill in Q2's 'pending CNV'

Three inputs: counts, annotations, gene positions

```r
library(infercnv)          # BiocManager::install("infercnv")
refs <- c("Immune cell", "Oligodendrocyte")
ann <- data.frame(row.names = colnames(gbm4),
                  group = ifelse(gbm4$celltype_author %in% refs,
                                 gbm4$celltype_author,
                                 paste0("obs_", gbm4$patient)))
write.table(ann, "data/infercnv_annot.txt", sep = "\t",
            col.names = FALSE, quote = FALSE)
# 基因座標檔：官方提供 hg38 gencode 版本（見腳本 05 的下載連結）
cts <- LayerData(gbm4, layer = "counts")
obj <- CreateInfercnvObject(raw_counts_matrix = cts,
                            annotations_file  = "data/infercnv_annot.txt",
                            gene_order_file   = "data/hg38_gencode_v27.txt",
                            ref_group_names   = refs)
obj <- infercnv::run(obj, cutoff = 1,        # Smart-seq 1；10x 0.1
                     out_dir = "output/infercnv",
                     cluster_by_groups = TRUE, denoise = TRUE, HMM = FALSE)
```

References must be cells known to be normal: immune and oligodendrocytes are safest. Not astrocytes — they may be the malignant ones | Script 05 §1 | Deeper: B16

How to read the inferCNV heatmap



Top: reference cells (flat). Bottom: each patient's malignant cells carry their own CNV set—the shared chr7+/chr10− is textbook GBM; the rest is private to each patient

chr7 gain and chr10 loss are textbook GBM, shared by all four; the rest are private to each patient — the genomic reason behind Q1's patient islands

Key inferCNV parameters: different platforms, different numbers

The commonest error is copying the cutoff from the wrong platform; the next is the wrong reference (Trap 1)

Parameter	Meaning	Smart-seq2	10x	Note
cutoff	Drop genes whose mean expression is below this	1	0.1	1 on 10x drops nearly all genes: a blank heatmap
ref_group_ names	Who is 'known normal'	Immune + Oligodendrocyte	same	≥ 50–100 cells per group never astrocytes
window_length	Smoothing window along the genome (genes)	101	101	Larger = smoother focal events vanish
HMM / denoise	Segment the signal into discrete CNV states	i6, optional	i6, optional	HMM is slow this course uses score + correlation
cluster_by_ groups	How observation cells are grouped in the plot	by patient	by sample	Per patient is what reveals private events

From heatmap to two numbers

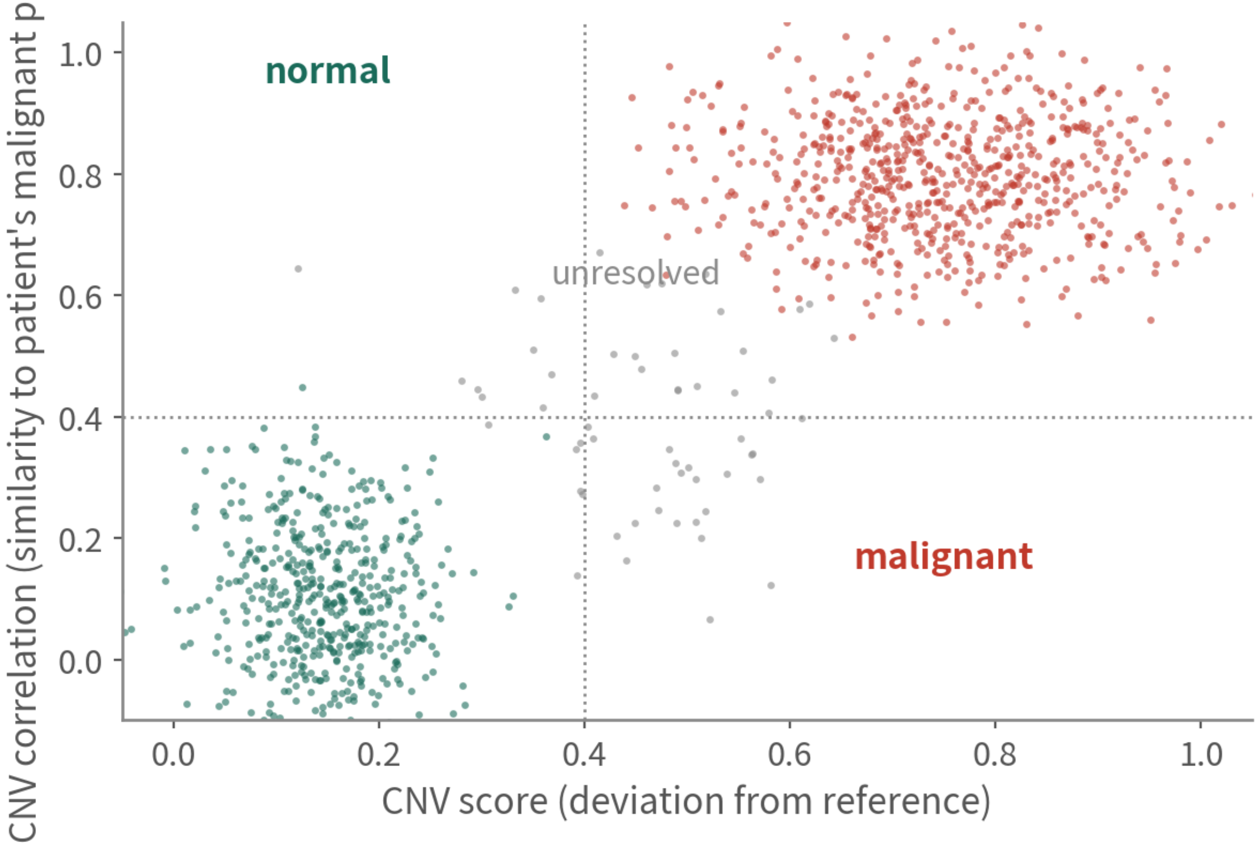
```r
d   <- "output/infercnv/"
cnv <- read.table(paste0(d, "infercnv.observations.txt"), check.names = FALSE)
ref <- read.table(paste0(d, "infercnv.references.txt"),   check.names = FALSE)
cnv.all <- cbind(cnv, ref)                          # 基因 × 細胞，已平滑、以參考為中心

cnv.score <- colMeans((cnv.all - 1)^2)                      # ① 偏離參考的程度
top <- names(sort(cnv.score, decreasing = TRUE))[1:200]
mal.profile <- rowMeans(cnv.all[, top])     # 最像惡性的 200 顆平均 profile
cnv.cor <- apply(cnv.all, 2, cor, y = mal.profile)  # ② 與惡性 profile 相關

gbm4$cnv.score <- cnv.score[colnames(gbm4)]
gbm4$cnv.cor   <- cnv.cor[colnames(gbm4)]
FeatureScatter(gbm4, "cnv.score", "cnv.cor", group.by = "celltype_author")
```

This is Tirosh 2016's approach: score plus correlation, plotted to see whether two clear groups emerge | Script 05 §2

Two numbers per cell, four quadrants

From heatmap to label: two numbers per cell

schematic / simulated



Triangulate: all three witnesses before you call it

① **CNV**

high score+high correlation with the patient's malignant profile

② **clustering**

patient-specific islands (normal cells mix)

③ **markers**

glial lineage (SOX2, OLIG2), not immune/vascular

into metadata

labels malignant/normal/unresolved —do not force the unsure ones

Top-right malignant, bottom-left normal, the middle unresolved. Do not force the unsure ones — reviewers prefer an honest grey zone

Triangulate, write to metadata, check the answer

```
gbm4$malignant <- with(gbm4@meta.data, ifelse(
  cnv.score > 0.02 & cnv.cor > 0.4, "malignant",           # 閾值看你的散點圖決定
  ifelse(cnv.score < 0.01 & cnv.cor < 0.2, "normal", "unresolved")))
# 第三個證人：譜系 marker。免疫 / 血管群若被標惡性，一律改 unresolved
gbm4$malignant[gbm4$malignant == "malignant" &
               gbm4$celltype_author %in% c("Immune cell", "Vascular")] <-
  "unresolved"
table(gbm4$malignant, gbm4$celltype_author)
```

```
             Neoplastic Astrocyte OPC Oligodendrocyte Immune cell Vascular Neuron
malignant         high       few  few           0            0         0       0
normal             few      most  most         all          all       all    most
unresolved          ...
```

Check against the authors' Neoplastic label: agreement is usually > 90%. The disagreements are worth a look — often periphery cells | Script 05 §3

Evidence of malignancy beyond CNV

CNV-quiet tumours (paediatric brain tumours, some blood cancers, near-diploid carcinomas) need different witnesses

Mutations and alleles

Smart-seq2 reads coding SNVs (IDH1 R132H, TP53); 10x catches them only near the 3′ end. HoneyBADGER and Numbat add allelic imbalance to CNV

A second CNV tool

copyKAT needs no reference; SCEVAN is automatic. Call malignant only where all agree; leave the rest unresolved

Expression programmes and pathology

High Neftel state scores, high proliferation, unlike any normal type; plus pathology (e.g. IDH1 R132H immunostain). Supporting, never sole evidence

04

Composition: what differs between core and periphery

Script 06a §1: proportions are compositional; the unit is the patient

Type proportions per patient and site

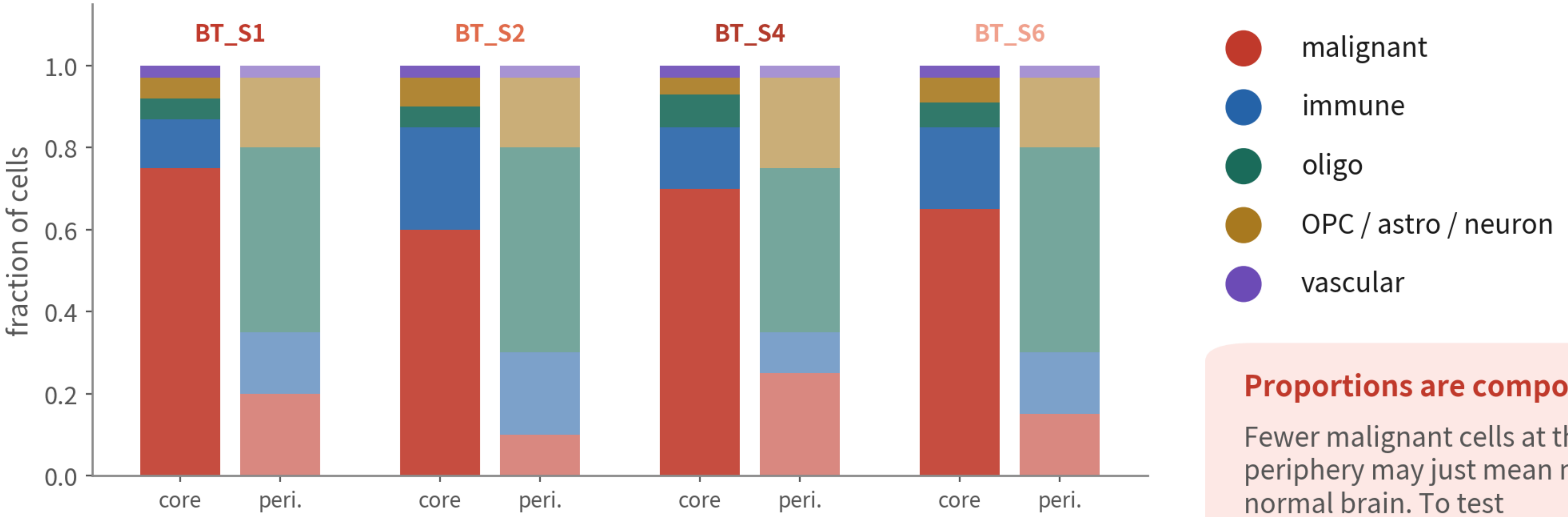
```r
gbm4$type <- ifelse(gbm4$malignant == "malignant",
                    "Malignant", gbm4$celltype_author)
prop <- gbm4@meta.data |> dplyr::count(patient, tissue, type) |>
        group_by(patient, tissue) |> mutate(frac = n / sum(n))
ggplot(prop, aes(tissue, frac, fill = type)) + geom_col() +
  facet_wrap(~ patient, nrow = 1) + theme_classic()

# 檢定：propeller（speckle 套件），以樣本為單位、配對設計
library(speckle)
propeller(clusters = gbm4$type, sample = paste(gbm4$patient, gbm4$tissue),
          group = gbm4$tissue, transform = "logit")
```

'Fewer malignant cells at the periphery' is tested with propeller / scCODA on n = 8 samples; a t-test on proportions is wrong | Deeper: B12

Composition: one pair per patient

Composition: cell-type proportions at core vs periphery per patient

schematic / simulated



Proportions are compositional

Fewer malignant cells at the periphery may just mean more normal brain. To test proportions use propeller/ scCODA, with patients as units

It counts only if all four patients agree. Proportions are closed: one type falling forces the others up

Three things about composition: closure, pairing, the unit

Closure

Proportions sum to 1: fewer malignant cells 'automatically' means more immune cells. Test on the logit or CLR scale (propeller, scCODA), never a t-test on raw proportions

Pairing

Core and periphery come from the same patient; a paired design removes between-patient variance and boosts power. In propeller, a ~ patient + tissue design matrix does it

The unit is the sample

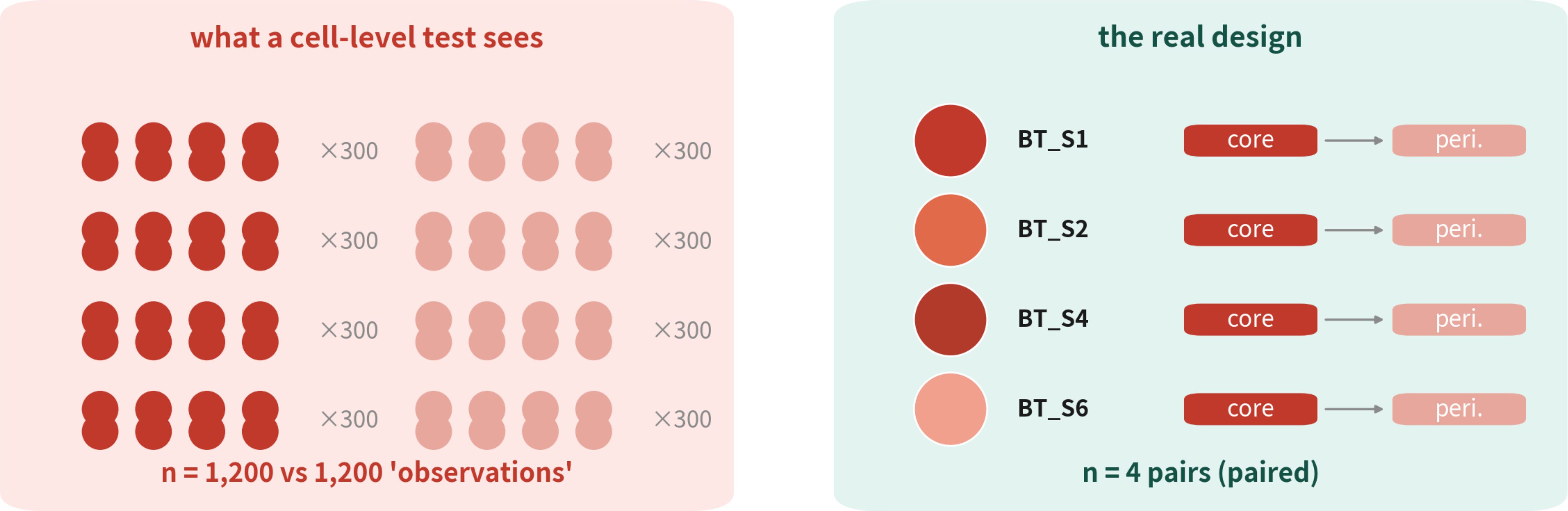
n = 8 samples (4 × 2), not 3,589 cells. A chi-square at the cell level gives absurdly small p values — the same mistake as Trap 1

05

DE: each cell type, core vs periphery

Scripts 06a (the pseudobulk mainline) and 06b (the cell-level fallback for too few samples)

Why a cell-level test hands you two thousand genes



Cells from one patient share genome and CNV—not independent.
Treating cells as observations=pseudoreplication; p values collapse

Malignant cells from one patient share one CNV set and are highly correlated. The real design is 4 pairs, not 2,400 observations

Pseudobulk in five steps

Pseudobulk: turn single-cell data back into one row per sample, then use mature bulk statistics

1. **malignant cells (CNV-called)** — one cell type only, or you compare composition
2. **sum by patient × site AggregateExpression** — sum raw RNA counts; the matrix must be integers
3. **an 8-column 'bulk' matrix** — 4 core+4 periphery, one sample per column
4. **DESeq2 ~ patient + site** — paired design: absorb patient first, then test site
5. **lfcShrink → results** — report and rank by shrunken log2FC

One-line check: sum(mat!=round(mat)) must be 0; otherwise you summed the data/scale.data/integrated layer

One cell type at a time — otherwise you are comparing composition (more normal brain at the periphery), not that type's change

Each cell type separately: why, and the thresholds

Mixed together = comparing composition

Core vs periphery differs in malignant cells, macrophages and oligodendrocytes in different ways. DE on all cells puts MBP and PLP1 on top — that is just more oligodendrocytes at the periphery

Two thresholds

MIN_CELLS = 20: a patient × site needs ≥ 20 cells to count as a pseudobulk sample; drop the cell otherwise. MIN_PAIRS = 2: at least two patients with both sites, or the paired design cannot exist

The output is one summary table

One DE table, one volcano and one enrichment set per type; de.summary lists pairs and significant genes per type. Malignant cells are the lead; the other types are context and control

Wrap pseudobulk + paired DESeq2 in a function; call it per type

```
pb_de <- function(obj, type, min.cells = 20, min.pairs = 2) {
  sub <- subset(obj, cells = colnames(obj)[obj$type == type])
  n   <- table(sub$patient, sub$tissue)
  ok  <- names(which(rowSums(n >= min.cells) == 2))   # 兩部位都夠的病人
  if (length(ok) < min.pairs) return(NULL)
  sub <- subset(sub, cells = colnames(sub)[sub$patient %in% ok])
  pb  <- as.matrix(AggregateExpression(sub, assays = "RNA",
                   group.by = c("patient", "tissue"))$RNA)
  stopifnot(sum(pb != round(pb)) == 0)                 # 一行抓雷：必須是整數
  coldata <- data.frame(patient = sub("_[^_]*$", "", colnames(pb)),
                        tissue  = factor(sub(".*_", "", colnames(pb)),
                                         levels = c("Periphery", "Tumor")))
  dds <- DESeq(DESeqDataSetFromMatrix(pb[rowSums(pb) >= 10, ], coldata,
                                      design = ~ patient + tissue))
  raw <- results(dds, name = "tissue_Tumor_vs_Periphery")    # stat → GSEA、火山圖
  shr <- lfcShrink(dds, coef = "tissue_Tumor_vs_Periphery", type = "ashr")
  list(res = data.frame(gene = rownames(raw), log2FC = raw$log2FoldChange,
                        log2FC_shrunk = shr$log2FoldChange, stat = raw$stat,
                        padj = raw$padj), pb = pb, coldata = coldata)
}
de <- lapply(types, function(ty) pb_de(gbm4, ty)); names(de) <- types
```

~ patient + tissue: the paired design absorbs patient differences first. Patient IDs contain underscores — split the column name at its last segment | Script 06a §2 | Deeper: B12

One DE result, three log2FCs: which goes where

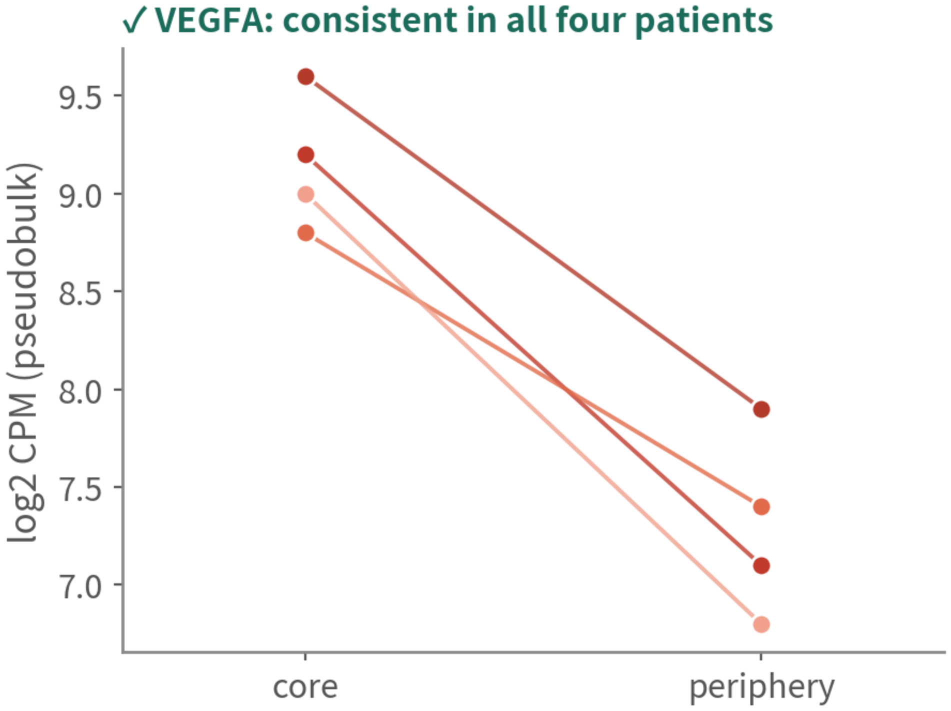
Shrinkage pulls the exaggerated FC of low-count, noisy genes toward 0; how hard it pulls differs a lot by method

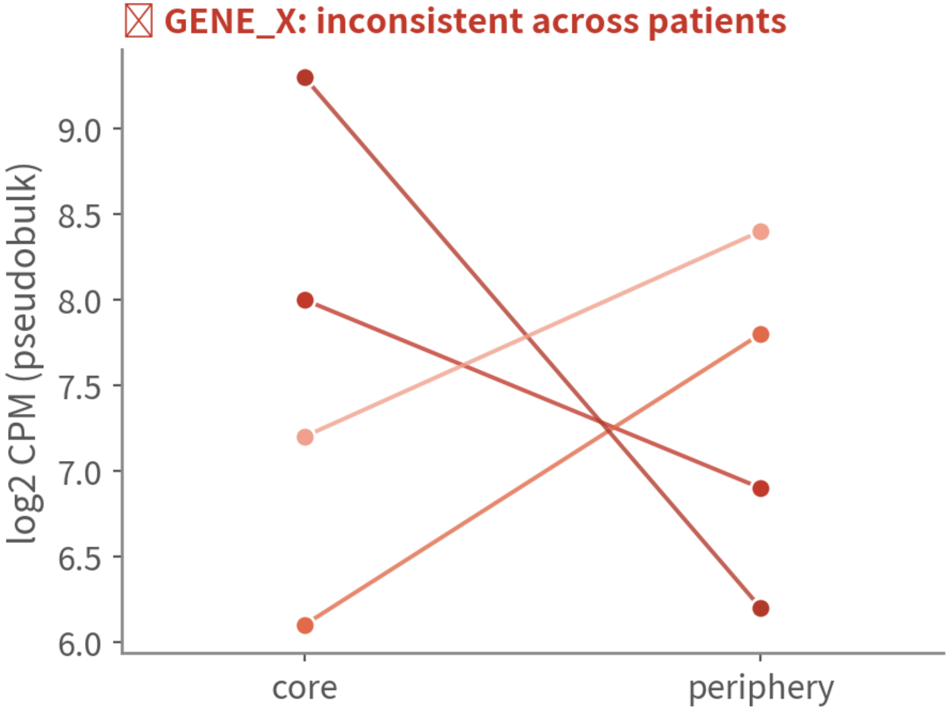
Column	Where it comes from	Behaviour	Use it for
raw log2FC	log2FoldChange from results(dds)	Low-count genes: huge FC, huge SE	The volcano x-axis (keeps the shape honest)
stat (Wald)	log2FC ÷ SE	Carries effect size and uncertainty; few ties	The GSEA ranking statistic
log2FC_shrunk (ashr)	lfcShrink(type = "ashr")	Stable at small n; keeps large effects, damps noise	Reported effect sizes; picking genes to validate
apeglm (not for small n)	lfcShrink(type = "apeglm")	At n = 8 the prior dominates: nearly every gene lands on one value → a bar-shaped volcano and all-tied GSEA ranks	Only with more samples (≥ 3 vs 3, ample counts)

When reporting, show it holds per patient

When reporting, show the difference holds per patient: one line per patient

schematic / simulated

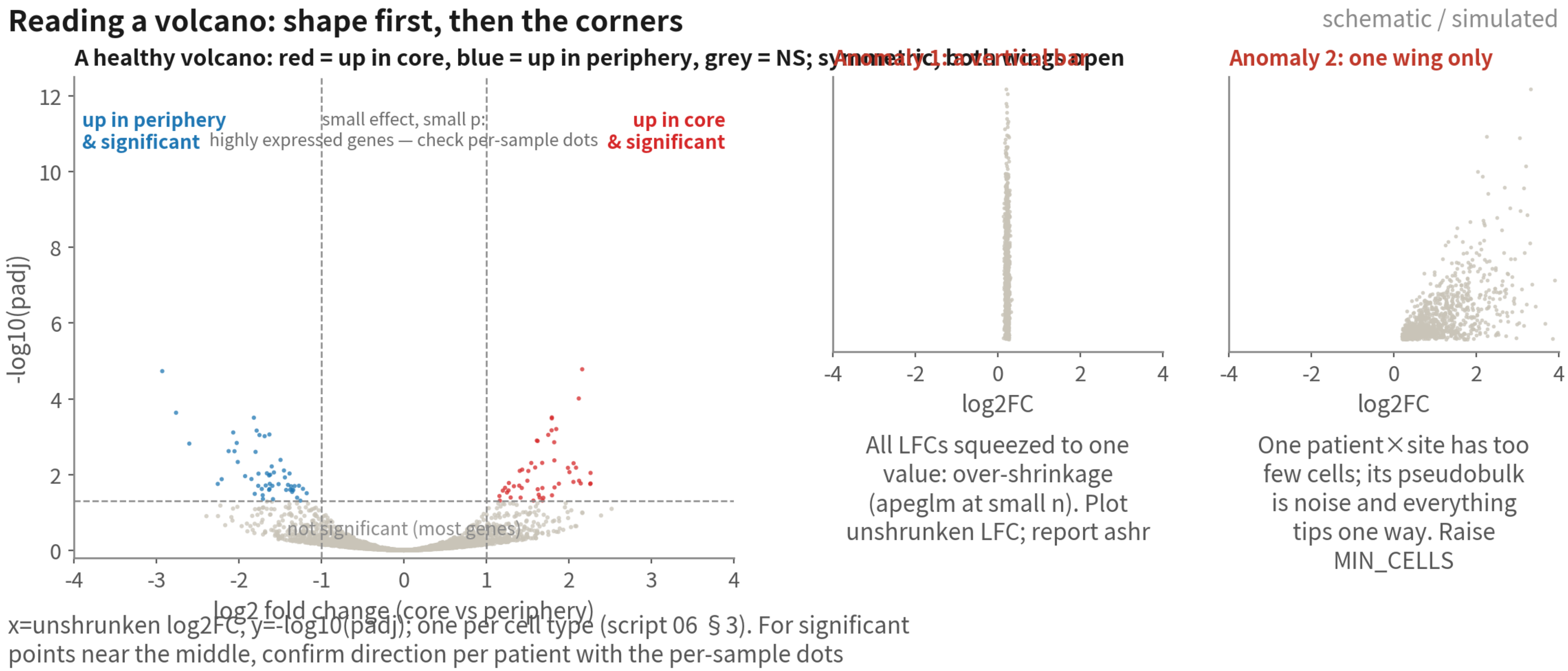




The left plot is more convincing than any volcano; the right one is still 'hugely significant' in a cell-level test

One line per patient. The left is more convincing than any volcano; the right is still 'hugely significant' at cell level

Reading a volcano: shape first, then the corners



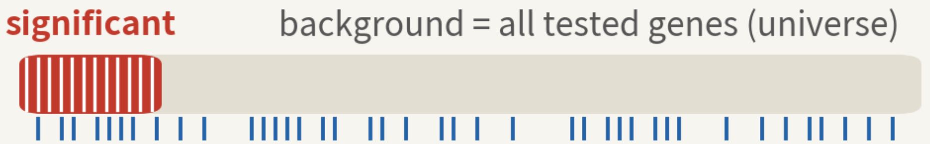
EnhancedVolcano: x = raw log2FC, y = -log10(padj), red / blue / grey = up in core / up in periphery / NS, one per type. A vertical bar = over-shrinkage; one wing only = a sample with too few cells | Script 06a §3

Same DE result, two enrichment questions: ORA counts, GSEA ranks

Same DE result, two enrichment questions: ORA counts shares, GSEA reads the ranking

schematic / simulated

ORA (over-representation): cut a threshold, then count



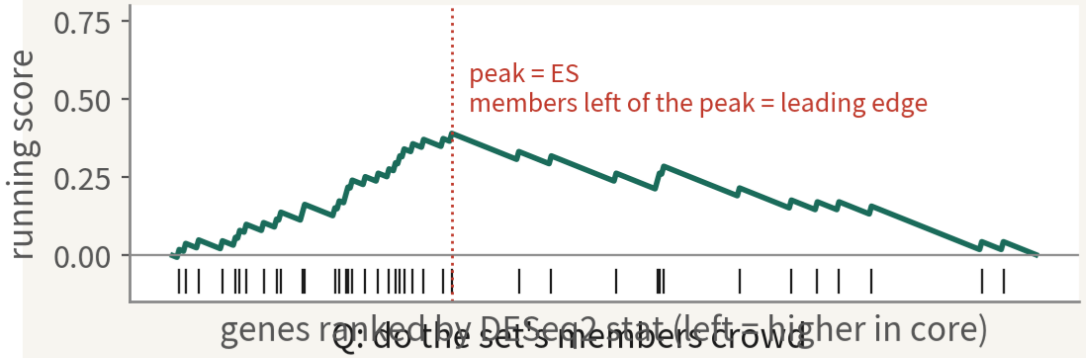
blue ticks = members of one gene set (e.g. GO 'response to hypoxia')

Q: is the share of blue ticks among significant genes higher than in the background?
hypergeometric / Fisher → padj, GeneRatio, Count

Needs: enough significant genes(≥ 10–20) and the right background. With a short DE list at small n, ORA has little to work with

Tools: clusterProfiler enrichGO/ enrichKEGG; plots: dotplot, cnetplot

GSEA: no threshold, use the whole ranked list



Q: do the set's members crowd toward one end of the list?
permutation test→NES (positive= higher in core), padj, leading edge

First choice at small n: every gene contributes evidence. Tool: fgsea (Hallmark/GO/KEGG alike)

Run both: GSEA as the main result, ORA up/down separately as a cross-check. Only write a pathway into the conclusion when both agree

GSEA uses every gene with no threshold — first choice at small n; ORA needs enough significant genes and the right background. Run both; a pathway goes into the conclusion only when both agree

Three gene-set databases: Hallmark, GO, KEGG

All from msigdbr and run with the same fgsea; ORA via clusterProfiler enrichGO / enrichKEGG

Database	Content and size	Strength	Caveat
Hallmark (H)	50 curated, minimally overlapping programmes: hypoxia, EMT, interferon, G2M…	Look here first: fits on one page, easiest to read, almost no redundancy	Only 50 sets; fine detail (which signal, which metabolic branch) is out of reach
GO BP (C5)	About 7,500 BP terms hierarchical, heavily overlapping	The finest grain; the default language of ORA; runs offline (org.Hs.eg.db)	One biology shows up as ten terms — simplify() first; terms with 3–5 hits stay out of the conclusion however significant
KEGG (C2 CP)	About 190 metabolic and signalling pathways, with pathway maps	Most intuitive for metabolism (glycolysis, OXPHOS); pathview draws on the map	enrichKEGG needs the internet; many are disease pathways ('Alzheimer disease') that are not mechanisms; the msigdbr copy is the legacy version

GSEA: three databases, every type, one loop

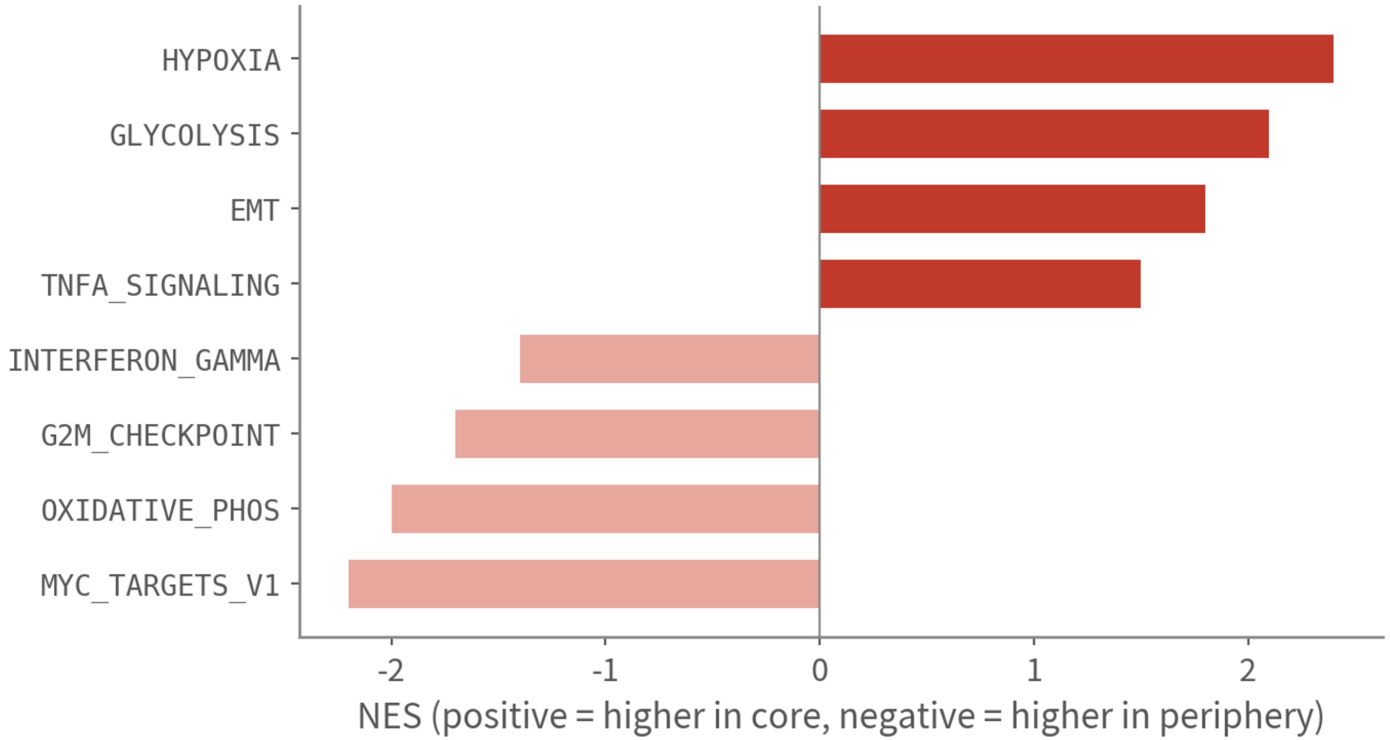
```
library(fgsea); library(msigdbr)
msig <- function(coll, sub = NULL) {
  m <- msigdbr(species = "Homo sapiens", collection = coll, subcollection = sub)
  split(m$gene_symbol, m$gs_name) }
gene.sets <- list(Hallmark = msig("H"), GO_BP = msig("C5", "GO:BP"),
                  KEGG = msig("C2", "CP:KEGG_LEGACY"))
rank_stat <- function(res) {                    # 排序用 Wald stat，不用 shrunken LFC
  r <- setNames(res$stat, res$gene); r <- r[!is.na(r)]
  sort(r + rnorm(length(r), sd = 1e-6), decreasing = TRUE) }   # 打散同分
run_gsea <- function(d, sets) rbindlist(lapply(names(sets), function(nm) {
  g <- fgsea(sets[[nm]], rank_stat(d$res), minSize = 15, maxSize = 500)   # eps 留預設
  g$collection <- nm; g$type <- d$type; g }))
gsea.all <- rbindlist(lapply(de, run_gsea, sets = gene.sets))
gsea.all[padj < 0.05][order(-abs(NES))][, .SD[1:5], by = .(type, collection)]
```

eps = 0 demands infinitely precise p values and runs for hours; the default 1e-10 is plenty. The background is automatically the tested genes | Script 06a §4 | Deeper: B13

GSEA: turn a hundred-odd genes into one sentence

GSEA (fgsea + Hallmark): turn 168 genes into one biological sentence

schematic / simulated



How to read

Core: hypoxia, glycolysis, EMT-like programmes—fits a necrotic centre. Periphery: proliferation and OXPHOS—infiltrating cells.

Sensible directions= the known-answer check passed; now read the leading-edge genes

Rank by the DESeq2 Wald stat (not the shrunken log2FC: massive ties) with no significance cut; the background is automatically the tested genes

Core hypoxic and glycolytic; periphery proliferative and OXPHOS — sensible directions pass the known-answer check. Now read the leading edge

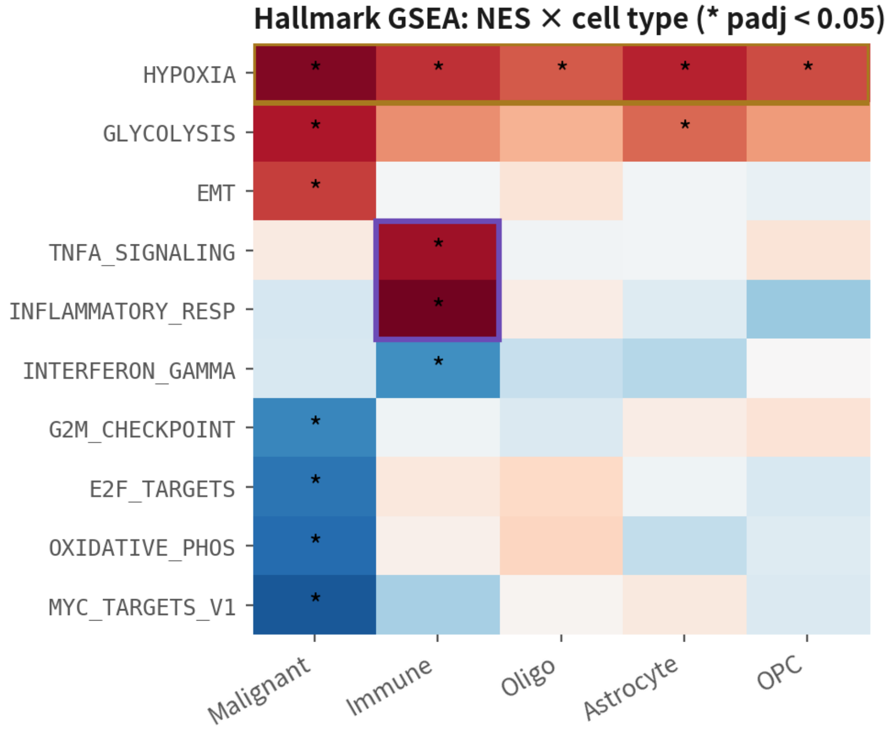
ORA: GO and KEGG in clusterProfiler, up and down separately

```
library(clusterProfiler); library(org.Hs.eg.db); library(enrichplot)
run_ora <- function(d, direction = "up", fc = 0.5, p = 0.05) {
  r   <- d$res[!is.na(d$res$padj), ]
  sig <- r$gene[r$padj < p & (if (direction == "up") r$log2FC > fc else r$log2FC < -fc)]
  if (length(sig) < 10) return(NULL)            # 顯著基因太少：改看 GSEA
  go <- enrichGO(sig, OrgDb = org.Hs.eg.db, keyType = "SYMBOL", ont = "BP",
                 universe = d$res$gene,          # ★ 背景 = 被檢定的基因，不是全基因組
                 minGSSize = 15, maxGSSize = 500) |>
        clusterProfiler::simplify(cutoff = 0.7)   # igraph 也有 simplify，寫全名
  ids  <- bitr(sig,        "SYMBOL", "ENTREZID", org.Hs.eg.db)
  uni  <- bitr(d$res$gene, "SYMBOL", "ENTREZID", org.Hs.eg.db)
  kegg <- enrichKEGG(ids$ENTREZID, organism = "hsa", universe = uni$ENTREZID) |>
          setReadable(org.Hs.eg.db, keyType = "ENTREZID")     # 要連網
  list(go = go, kegg = kegg) }
ora <- run_ora(de[["Malignant"]], "up")
dotplot(ora$go, showCategory = 12); dotplot(ora$kegg, showCategory = 12)
cnetplot(ora$go, showCategory = 5)              # 詞條–基因網路：哪些基因撐起哪個詞條
```

Leave universe empty and clusterProfiler uses the whole genome as background — twice the terms, all spurious. Up and down separately, because 'hypoxia up' and 'proliferation down' cancel when mixed | Script 06a §4

Reading enrichment: the NES heatmap and the dotplot

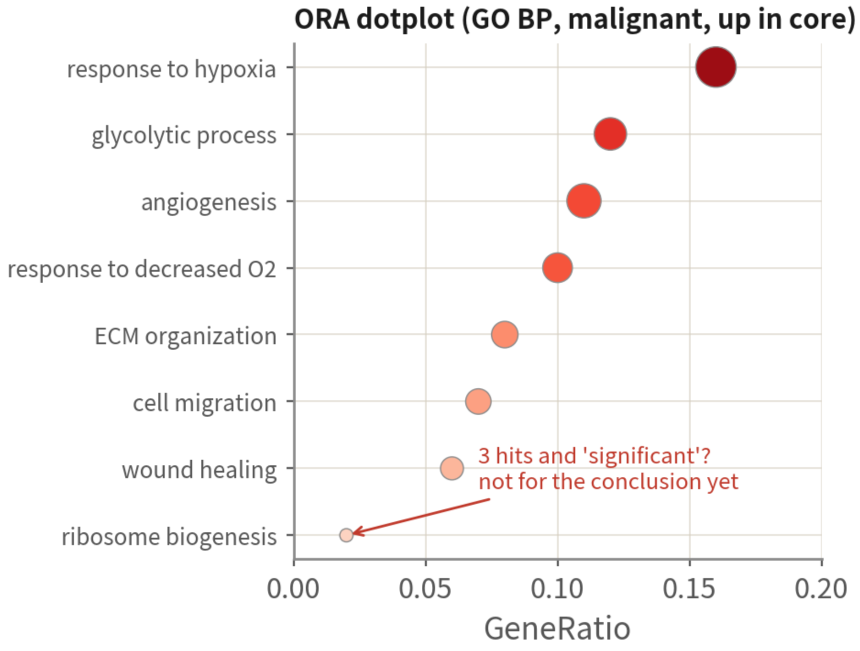
Reading enrichment: the NES heatmap says whose pathway, the dotplot says which terms to trust schematic / simulated



whole row red: a microenvironment programme (hypoxia)—every cell type feels it

one column only: a cell-intrinsic programme (inflammation in immune, proliferation in malignant)

Tells you at a glance whose pathway it is—a question a mixed-cells DE cannot even ask



x=share of significant genes in the term; size=hits; colour=padj. Read only the top few with high GeneRatio and small padj

A whole red row = a microenvironment programme every type feels; one column = a cell-intrinsic programme. In the dotplot read only the top few with high GeneRatio and small padj | Script 06a §4, plots A–C

The five most common enrichment mistakes

Wrong background

The ORA universe must be the tested genes (expressed, entered into DESeq2), not the genome. Single cell detects ten-odd thousand genes; a twenty-thousand background inflates every term

Direction flipped or mixed

The sign of NES follows the tissue level order; unsplit ORA lets opposite programmes cancel. Confirm direction with the known answer (core should be hypoxic) before reading on

Redundancy is not evidence

GO's 'response to hypoxia', 'response to decreased oxygen levels' and 'response to oxygen levels' are one thing. Count after simplify(); report one term, not three

Concluding from too few hits

Terms with 3–5 hits stay out of the conclusion however significant; a leading edge needs a dozen genes or more, each of which you can explain

Forgetting whose pathway it is

Hypoxia rises in malignant cells and in macrophages alike — that is the microenvironment, not a malignant trait. Per-type analysis and a NES heatmap separate the shared from the specific

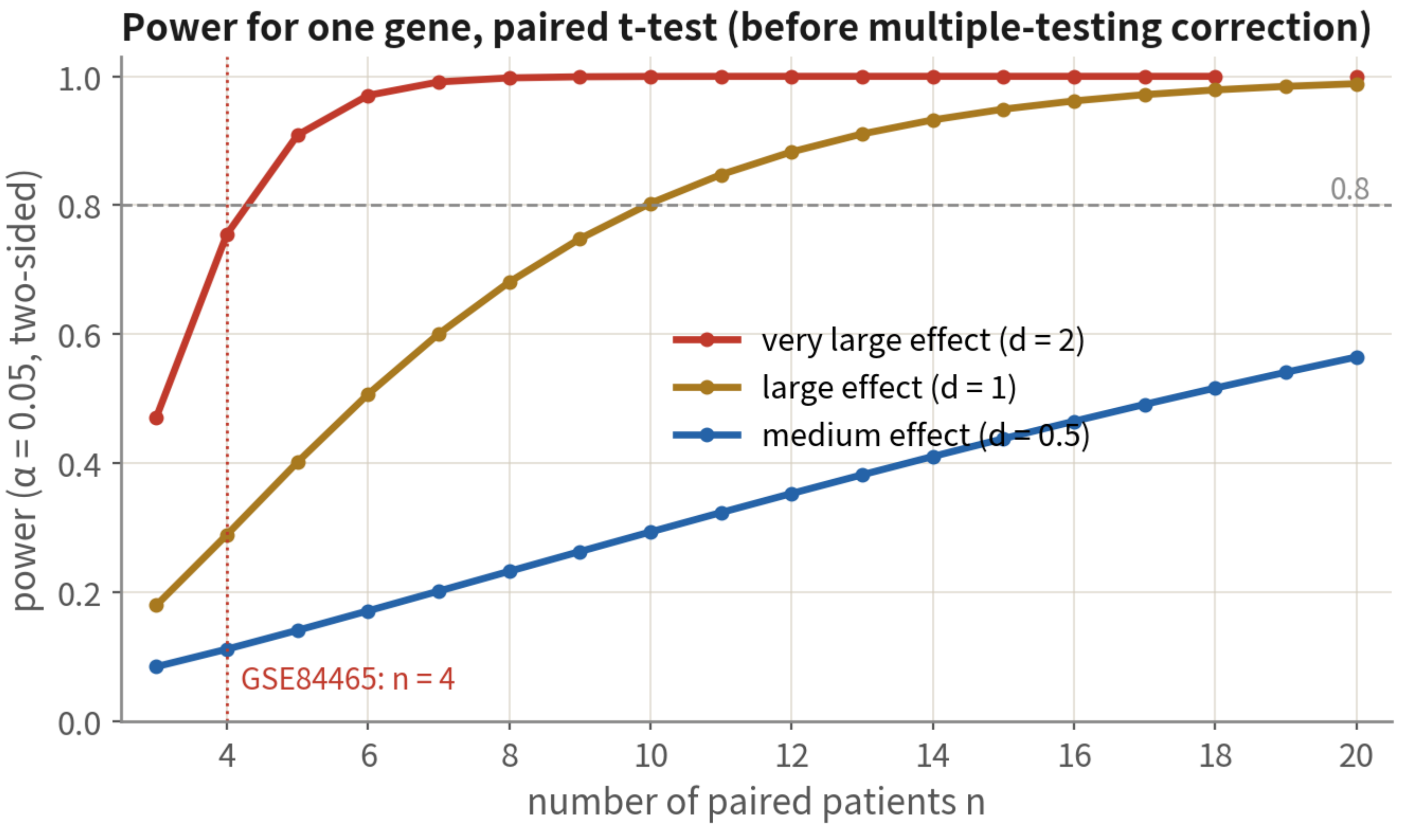
Deliverables: per-type DE tables, volcanoes, GSEA and ORA summaries

```
for (ty in names(de))                                            # 每種型別一份表
  write.csv(de[[ty]]$res, sprintf("output/06_de_%s.csv", ty), row.names = FALSE)
write.csv(de.summary, "output/06_de_summary_by_type.csv")   # 幾對病人、幾個顯著基因

vol <- lapply(de, function(d) {                                  # 紅 = 核心高、藍 = 邊緣高、灰 = NS
  kv <- with(d$res, ifelse(padj < 0.05 & log2FC > 1, "#D62728",
                  ifelse(padj < 0.05 & log2FC < -1, "#1F77B4", "grey70")))
  names(kv) <- c("#D62728" = "Up in core", "#1F77B4" = "Up in periphery", grey70 = "NS")[kv]
  EnhancedVolcano(d$res, lab = d$res$gene, x = "log2FC", y = "padj", colCustom = kv,
                 pCutoff = 0.05, FCcutoff = 1, title = paste0(d$type, ": core vs periphery")) })
ggsave("output/figs/06_volcano_all_types.png", wrap_plots(vol, ncol = 2), bg = "white")

gsea.all[, leadingEdge := sapply(leadingEdge, paste, collapse = ";")]   # list 欄轉字串
write.csv(gsea.all, "output/06_gsea_all_types.csv", row.names = FALSE)
# NES 熱圖（列 = pathway、欄 = 型別）與 ORA dotplot：見腳本 06a §4 圖 A–C
plotEnrichment(gene.sets$Hallmark[["HALLMARK_HYPOXIA"]], rank_stat(de$Malignant$res))
```

Tables carry raw and shrunken LFC, padj and baseMean; the volcano plots padj, not p; the GSEA table includes the leading edge. Everything regenerates from the script; nothing in output is hand-made | Script 06a §5

How many patients are enough? Power comes from patients

How many patients are enough? Power comes from patients, not cells



How to read this

1. n=4 catches only very large effects; medium effects have<20% power
2. add multiple-testing over thousands of genes and fewer survive FDR—a short DE list at n=4 is expected, not a mistake
3. to see medium effects, a paired design needs roughly≥10 patients; unpaired needs more
4. when power is short: treat n =4 as hypothesis-generating and validate in public or bulk cohorts

n = 4 sees only very large effects, so a short DE list is expected. Treat n = 4 as hypothesis-generating and validate in public cohorts — Trap 3 stated positively

When pseudobulk is impossible: three lines of defence

First admit where you are

Pseudobulk needs ≥ 2–3 biological replicates per condition. Too few patients (skipped in 06a's de.summary, like OPC here) means falling back to cell level — where cell counts inflate p, so the result can only be hypothesis-generating

Defences 1 + 2: covariate and per-patient consistency

Run Seurat's default Wilcoxon as the baseline, then MAST with patient in latent.vars (Wilcoxon cannot) — genes significant only under Wilcoxon are mostly patient effects. Then compute logFC per patient and keep one-direction genes; NA in the table = that patient has < 5 cells on one side

Defence 3: label permutation

Shuffle tissue labels within patients and rerun: real signal should far exceed the 'significant' count under permutation. If they are comparable, that is pseudoreplication talking. State method and limits on the deliverable table

Cell-level DE done properly (script 06b)

```
obj <- subset(gbm4, cells = colnames(gbm4)[gbm4$type == "OPC"])
table(obj$patient, obj$tissue)          # 先看樣本結構：兩部位都夠的病人只有 1 位
Idents(obj) <- "tissue"
de.wilcox <- FindMarkers(obj, ident.1 = "Tumor", ident.2 = "Periphery",
                         logfc.threshold = 0.25)   # Seurat 預設 Wilcoxon：當基準
de <- FindMarkers(obj, ident.1 = "Tumor", ident.2 = "Periphery",
                  test.use = "MAST", latent.vars = "patient",   # 防線一：病人當共變量
                  logfc.threshold = 0.25, min.pct = 0.1)
setdiff(head(de.wilcox$gene, 30), head(de$gene, 30))  # 只在預設法顯著 ≈ 病人效應
# 防線二：逐病人 logFC，方向一致才留（單邊 < 5 顆的病人不算）
# 防線三：tissue 在病人內置換 20 次，比較「顯著基因數」的真實值 vs 置換分布
perm.sig <- replicate(20, {
  o <- obj; o$tissue <- ave(as.character(o$tissue), o$patient, FUN = sample)
  Idents(o) <- o$tissue
  sum(FindMarkers(o, ident.1 = "Tumor", ident.2 = "Periphery",
                  logfc.threshold = 0.25)$p_val_adj < 0.05, na.rm = TRUE) })
de$method <- "cell-level MAST; hypothesis-generating"   # 交付表明寫限制
```

The point is not the p value — it is the three defences and honest framing. Whenever pseudobulk is possible, use 06a; 06b is the fallback when the sample structure forbids it | Script 06b

06

After DE: choosing the road

Six roads tumour studies take; each needs the right data

Six roads and what each needs

After DE: six roads tumour studies take, and the data each needs

cell communication

malignant↔TAM ligand–receptor axes (e.g. SPP1, CSF1)

needs: reliable labels; multiple types

`CellChat`

malignant states & plasticity

MES/AC/OPC/NPC-like scores, cycling

needs: CNV-called malignant cells+published gene sets

`AddModuleScore`

trajectory

state transitions within malignant cells (not between types)

needs: a sampled continuous process; a justified root

`Slingshot`

CNV subclones

subclonal structure within a patient

needs: inferCNV HMM or CopyKAT

`inferCNV`

deconvolving bulk cohorts

use a single-cell reference on TCGA-GBM, link to survival

needs: same-tissue reference+bulk

`MuSiC / BayesPrism`

spatial

which malignant state sits near vessels/necrosis

needs: Visium / Xenium data

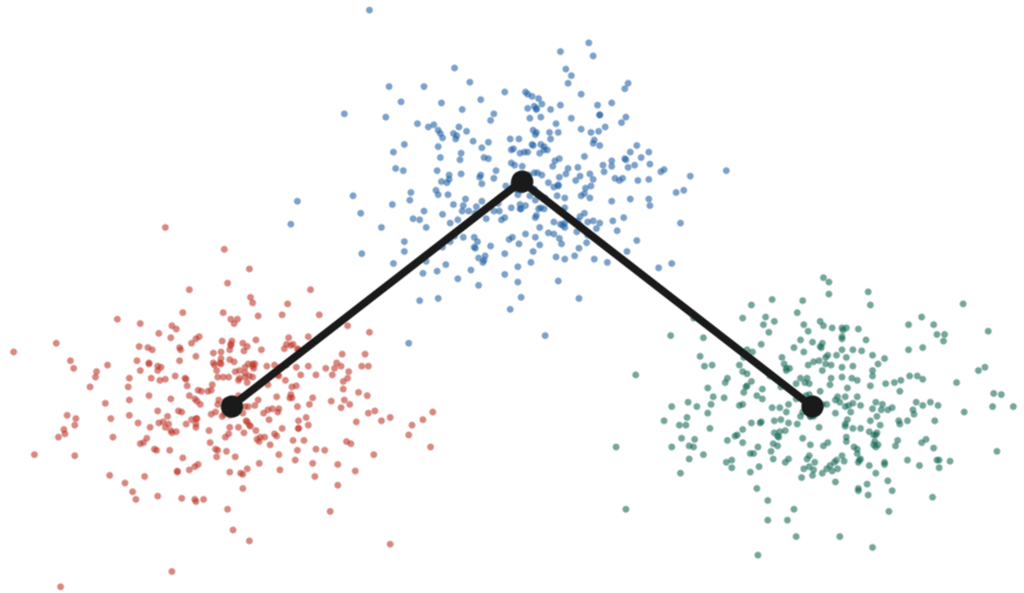
`Seurat spatial`

Check the 'needs' line first, then choose. Pick the wrong data and the whole road is void

The most common downstream misuse

schematic / simulated

A 'differentiation path' malignant → immune → oligo?



A trajectory tool always draws a line

- ✗ Malignant cells do not become macrophages. The line does not exist
- ✗ Branch points at type boundaries are the classic artefact
- ✓ A sensible tumour trajectory: state transitions inside malignant cells (OPC-like→AC-like), after CNV calling, per patient
- ✓ A snapshot gives no direction; the root needs independent evidence (cycling, stemness, velocity)

Only one kind of trajectory makes sense in a tumour: state transitions inside malignant cells, after CNV calling, per patient

07

Cell–cell communication: CellChat from run to reading

Script 07: six plots, each answering one question

Cell–cell communication (CellChat / LIANA): three preconditions

The favourite downstream analysis in tumour studies — and the most often misused

Within one sample

Ligand and receptor must come from the same site of the same patient — macrophages from patient A cannot talk to tumour cells of patient B. Run per sample, then compare

Both ends need enough cells

Default thresholds are often 10% of cells; ambient RNA fabricates ligands. Run after SoupX, and only with CNV-called malignant cells

Results are hypotheses

Ligand–receptor co-expression at the RNA level is not binding and not signalling. Write 'X drives Y' only with a second line of evidence (spatial, protein, perturbation)

What CellChat computes

CellChatDB: ligand–receptor pairs (~3,300 in human)

Secreted signalling
cytokines, growth factors:
SPP1–CD44, TGFB1–TGFBR

ECM–receptor
matrix: COL1A1–integrins, laminins

Cell–cell contact
requires adjacency:
PD-L1–PD-1, NOTCH–JAG, MHC–TCR

Multi-subunit receptors (ITGAV+ITGB1) and co-factors are encoded in the database—the difference from simply multiplying two genes

How a communication probability is computed

① per group, take the trimean of ligand L and receptor R (robust; needs≥25% of cells expressing)

② mass action: P=L・R/(K+L・R); geometric mean over subunits; agonists multiply, antagonists divide

③ permutation: shuffle labels 100×, recompute P; keep pairs whose real P beats 95% of shuffles (p<0.05)

④ sum L–R pairs of one pathway (e.g. 3 pairs in SPP1)→pathway level; sum all pathways→the aggregated network

Three output levels: L–R pair→pathway→group-to-group. Every plot that follows is a slice of one level

The database decides which pairs are possible, expression decides which hold, permutation decides which are not chance. Three output levels: L–R pair, pathway, group-to-group

One CellChat run: Seurat object to aggregated network (once per sample)

```
library(CellChat)
run_cc <- function(obj, label = "cc_label") {
  obj$samples <- factor(paste(obj$patient, obj$tissue))       # v2 要求 samples 欄
  cc <- createCellChat(obj, group.by = label, assay = "RNA")   # 用 data 層
  cc@DB <- subsetDB(CellChatDB.human, search = "Secreted Signaling")
  cc <- subsetData(cc) |> identifyOverExpressedGenes() |>
        identifyOverExpressedInteractions()
  cc <- computeCommunProb(cc, type = "triMean",     # 注意大小寫
                          population.size = TRUE)         # 群大小校正
  cc <- filterCommunication(cc, min.cells = 20)           # < 20 顆的群不參與
  cc <- computeCommunProbPathway(cc) |> aggregateNet()
  netAnalysis_computeCentrality(cc)
}
core <- run_cc(subset(gbm4, tissue == "Tumor" & patient == "BT_S2"))
peri <- run_cc(subset(gbm4, tissue == "Periphery" & patient == "BT_S2"))
# 4 位病人 × 2 部位 = 8 個物件；lapply 跑完存成 list
```

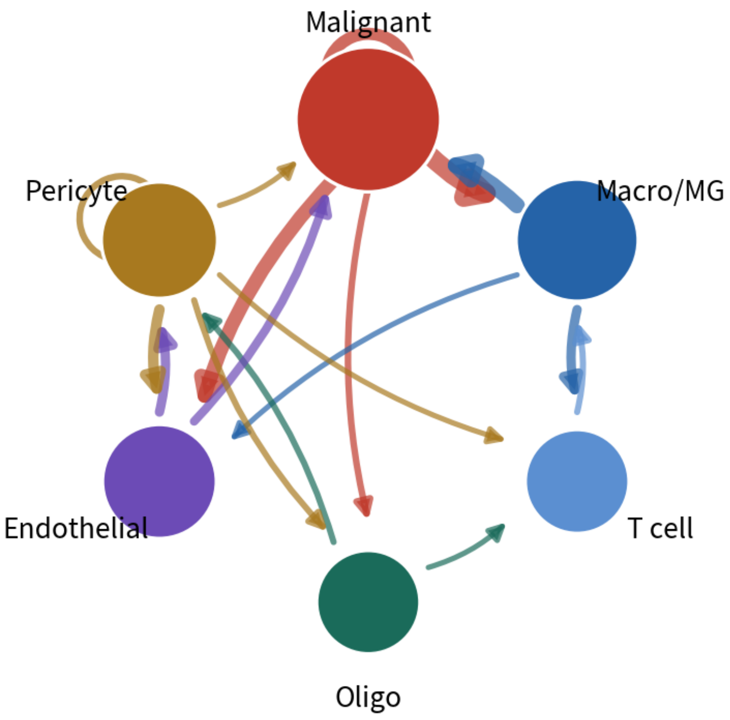
Three keys: the log-normalized data layer, one run per sample, labels after CNV calling. About 5–15 min per sample | Script 07 §1

Plot 1: netVisual_circle — the whole network at a glance

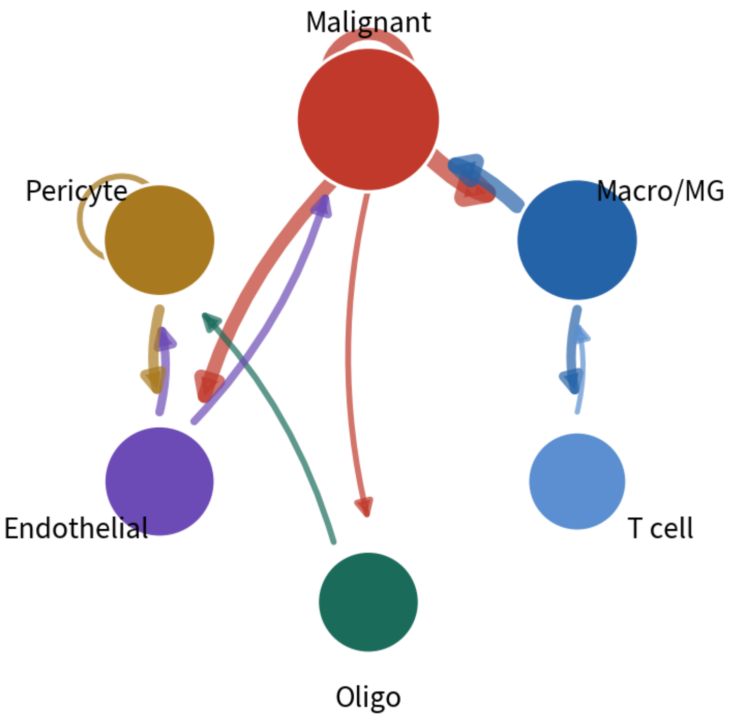
Plot 1: netVisual_circle — the bird's-eye view of the network

schematic / simulated

Number of interactions



Interaction weights / strength



How to read

One node per group; node size=total outgoing

Edge colour=sender; the arrow points at the receiver

Edge width=number of significant L–R pairs (left) or summed probability (right)

Left and right often disagree: many pairs≠strong. Find the main axis on the right, then return to the left

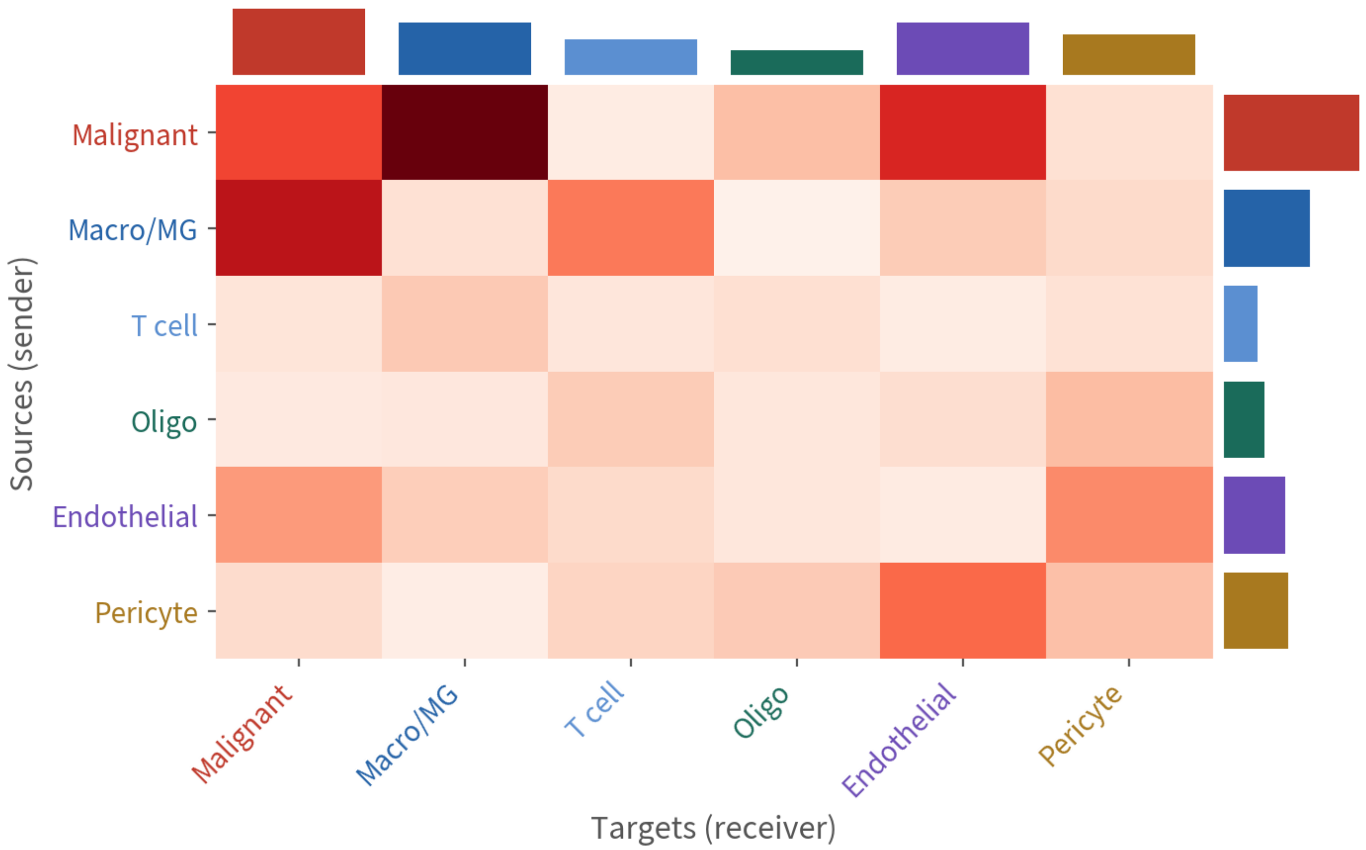
Self-loop=within a group (malignant autocrine)

groupSize <- table(cc@idents); netVisual_circle(cc@net$count, vertex.weight = groupSize), and again with cc@net$weight. Find the main axis on the right first

Plot 2: netVisual_heatmap — who sends to whom

Plot 2: netVisual_heatmap — who sends to whom

schematic / simulated



How to read

Rows=senders, columns= receivers; darker=stronger

Right bars=row sums (top senders); top bars=column sums (top receivers)

Typical GBM: malignant↔macrophage is the brightest pair; malignant→endothelial (VEGF) is often bright too

Better than the circle plot for values; the circle plot is better for direction

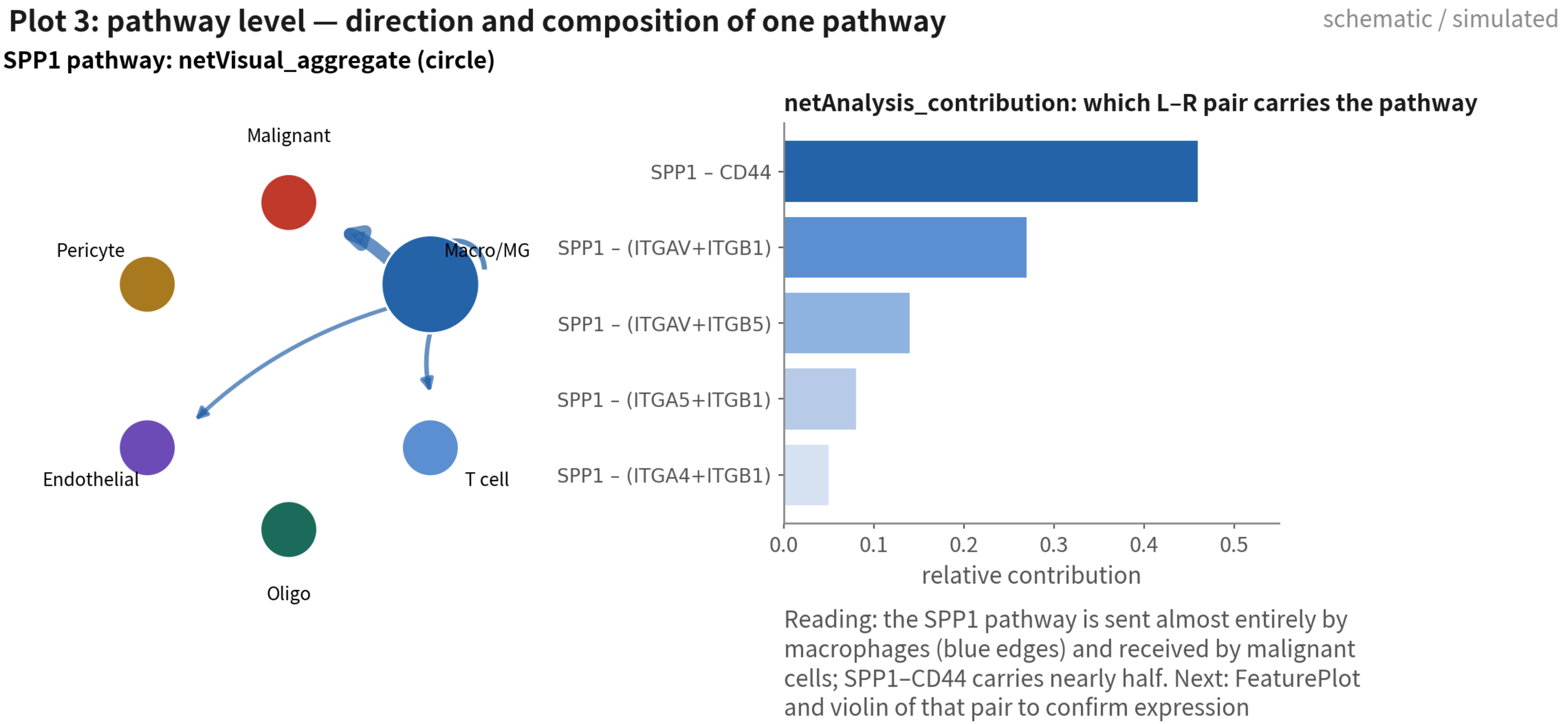
netVisual_heatmap(cc, measure = "weight", color.heatmap = "Reds"). Rows are sources, columns targets; bars are sums. Quote numbers from this one, read direction from the circle

Pathway level: rank first, then take one pathway apart

```
cc@netP$pathways                                    # 顯著的路徑清單
netAnalysis_signalingRole_heatmap(cc, pattern = "outgoing", height = 8)   # 圖 4 左
netAnalysis_signalingRole_heatmap(cc, pattern = "incoming", height = 8)   # 圖 4 右
netAnalysis_signalingRole_scatter(cc)                                     # 圖 4 散點

pw <- "SPP1"
netVisual_aggregate(cc, signaling = pw, layout = "circle")   # 圖 3 左（也可 "chord"、"hierarchy"）
netAnalysis_contribution(cc, signaling = pw)                 # 圖 3 右：哪對 L–R 撐起路徑
netVisual_heatmap(cc, signaling = pw, color.heatmap = "Reds")# 單一路徑的來源 × 目標
plotGeneExpression(cc, signaling = pw)                       # 該路徑所有基因的 violin：驗證表現
netAnalysis_signalingRole_network(cc, signaling = pw)        # sender / receiver / mediator / influencer
```

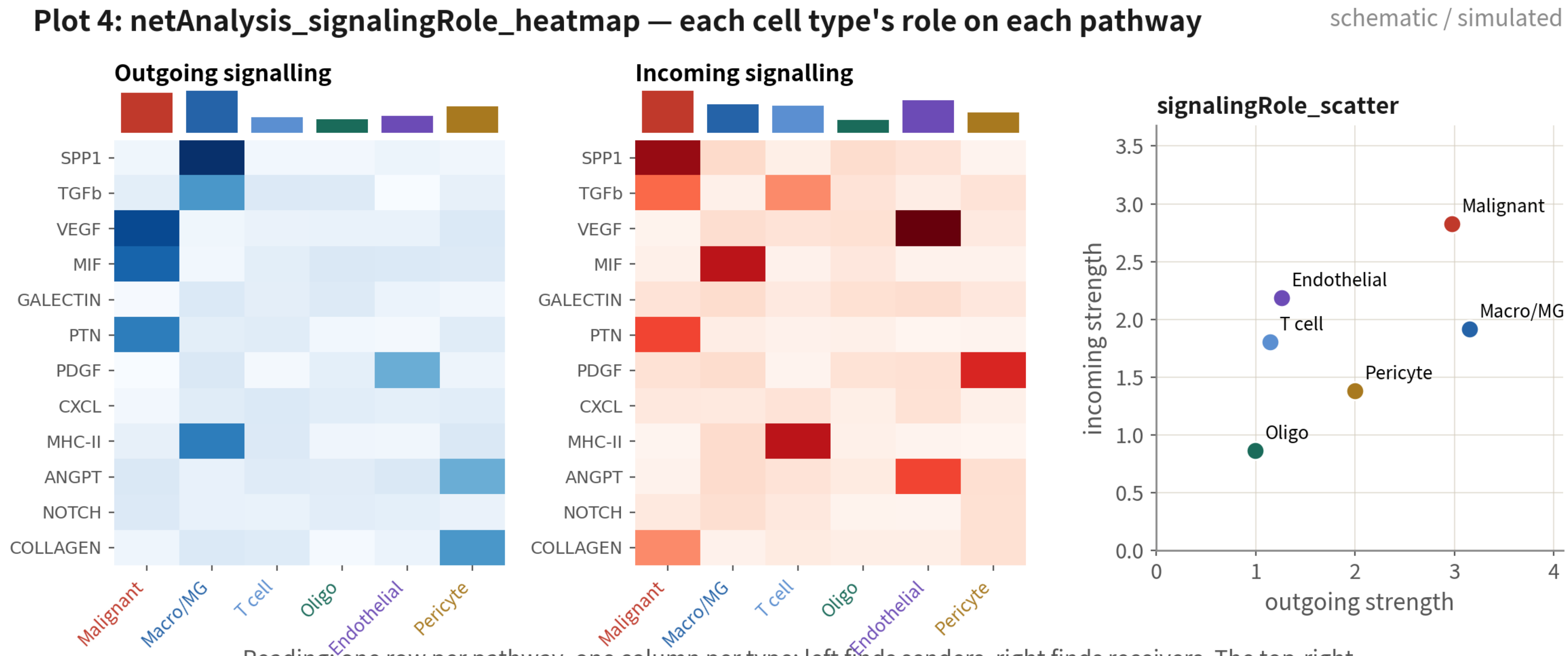
Reading order: signalingRole for the actors → aggregate for direction → contribution for the concrete pair → plotGeneExpression to verify. Pathway names from cc@netP$pathways | Script 07 §2

Plot 3: direction and composition of one pathway



SPP1 is the textbook TAM → malignant signal in GBM. Seeing this, the next step is not the paper — it is the violin of SPP1 and CD44 to confirm expression and fraction

Plot 4: signalingRole — each type's role on each pathway



Reading: one row per pathway, one column per type; left finds senders, right finds receivers. The top-right of the scatter is the hub that both sends and receives (in GBM usually malignant cells and macrophages)

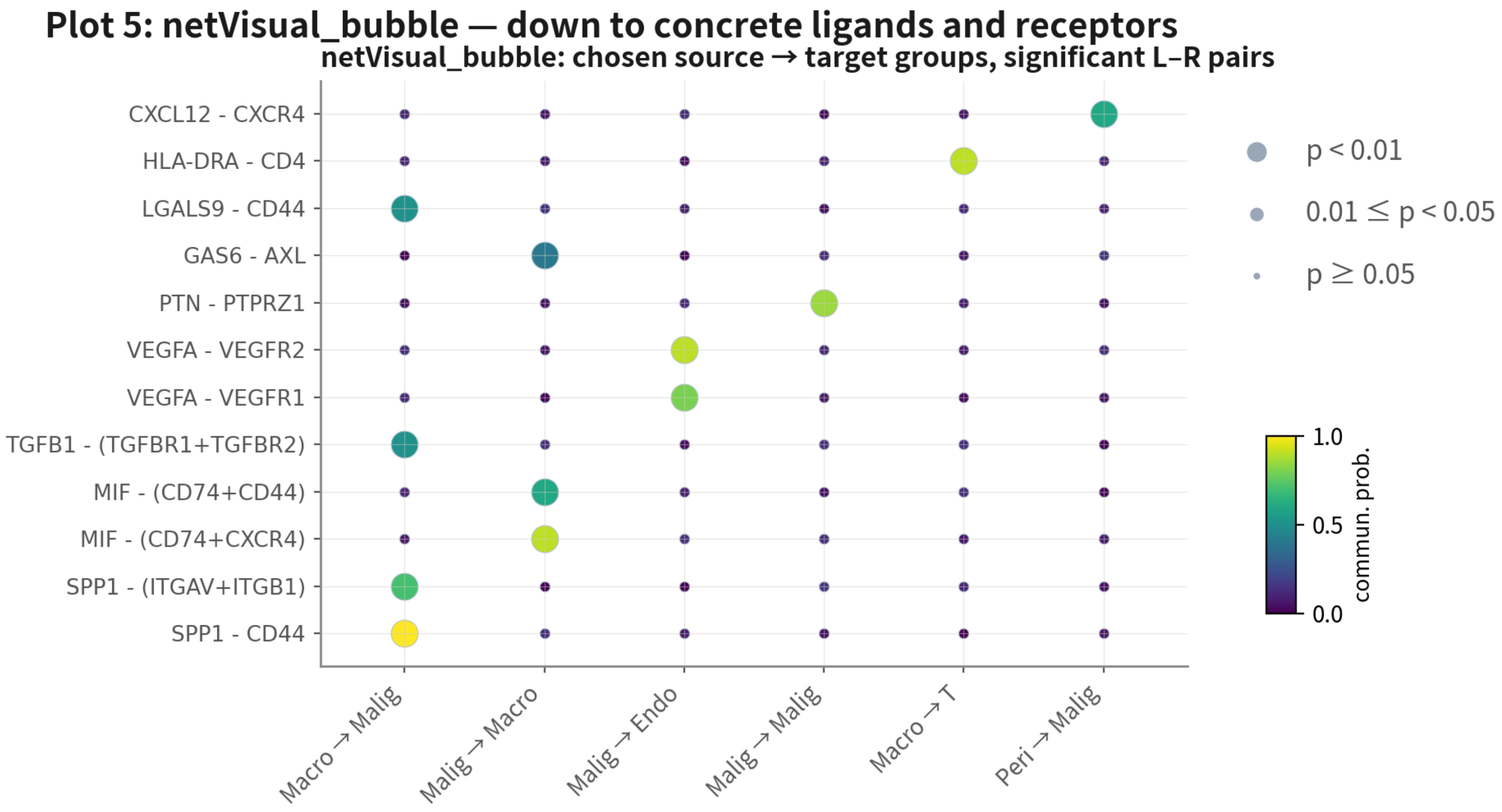
The most informative plot in the analysis: malignant cells send VEGF / MIF / PTN, macrophages send SPP1 / TGFb / MHC-II, at one glance. The top-right of the scatter is the hub

Down to concrete ligand–receptor pairs, then compare two conditions

```
# 圖 5：指定來源與目標群，列出顯著的 L-R 對
netVisual_bubble(cc, sources.use = c("Macro/MG", "Malignant"),
                 targets.use = c("Malignant", "Macro/MG", "Vascular"),
                 remove.isolate = TRUE)
sig3 <- head(intersect(c("SPP1", "MIF", "VEGF"),
             cc@netP$pathways), 3)              # 名稱以 cc@netP$pathways 為準
netVisual_bubble(cc, signaling = sig3)           # 只看幾條路徑

# 圖 6：兩條件比較——先合併
cc.m <- mergeCellChat(list(Core = core, Periphery = peri),
                      add.names = c("Core", "Periphery"))
compareInteractions(cc.m, group = c(1, 2))                    # 互動數
compareInteractions(cc.m, group = c(1, 2), measure = "weight")
netVisual_diffInteraction(cc.m, weight.scale = TRUE)          # 差異 circle：紅增藍減
netVisual_heatmap(cc.m, measure = "weight")                   # 差異熱圖
rankNet(cc.m, mode = "comparison", stacked = TRUE, do.stat = TRUE)  # Wilcoxon
netVisual_bubble(cc.m, sources.use = "Macro/MG", targets.use = "Malignant",
                 comparison = c(1, 2), angle.x = 45)          # 同一對 L-R 在兩條件
```

Each of the four patients has a pair of objects; when comparing, look for pathways consistent across all four, not one patient. The unit is still the patient | Script 07 §3–4

Plot 5: netVisual_bubble — the one that goes into the paper



schematic / simulated

How to read

One L–R pair per row; one source→ target per column

Colour=probability, size=p: chase only the big and yellow

One ligand hitting several receptors (SPP1→CD44/integrins) fills several rows— they share the same ligand expression

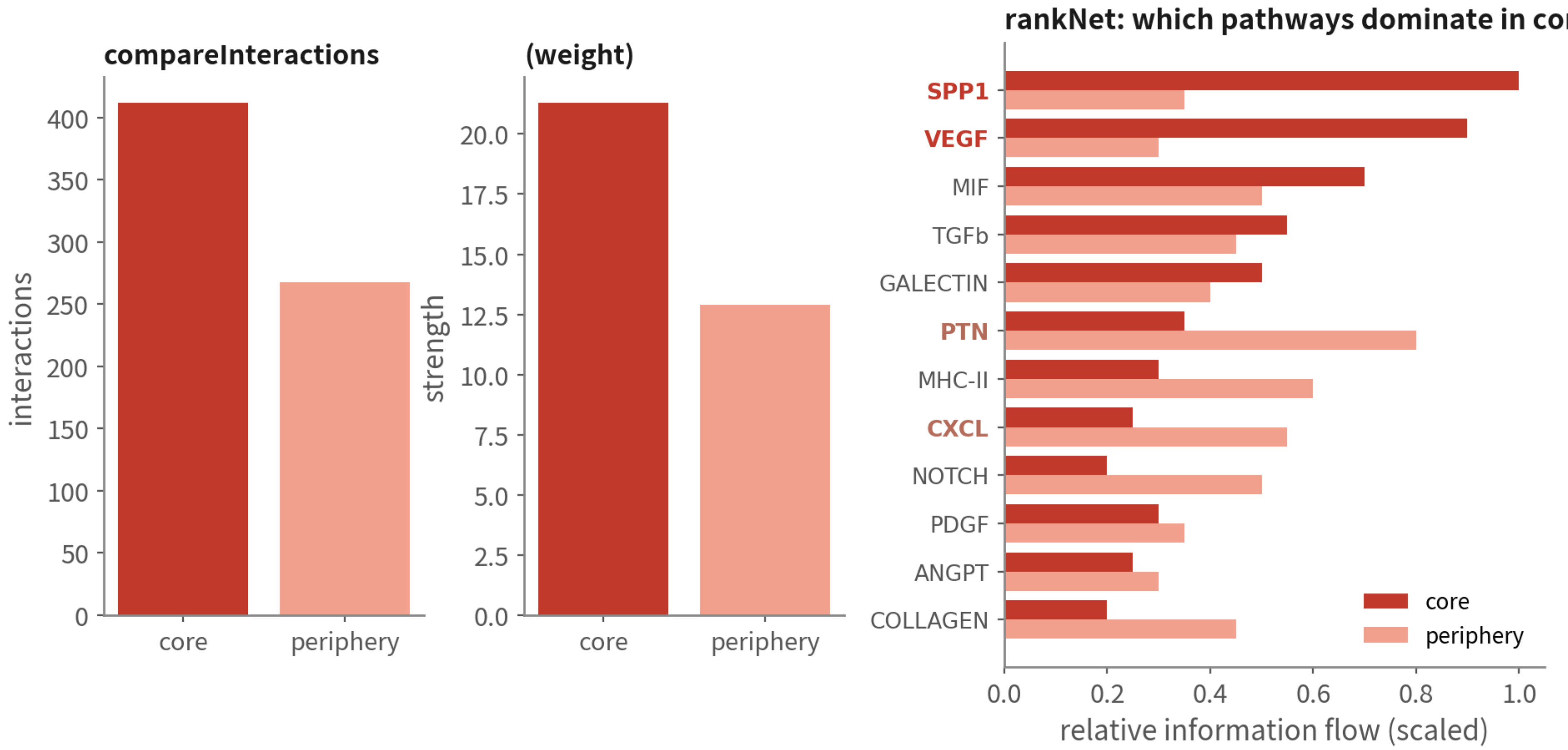
This is the plot that goes into the paper

Chase only the big and yellow; one ligand hitting several receptors fills several rows. Once you have a concrete pair, go back to Seurat for FeaturePlot and VlnPlot

Plot 6: core vs periphery — after mergeCellChat

Plot 6: comparing two conditions — three plots after mergeCellChat

schematic / simulated



Reading

Run per patient, then merge and compare; bold= significant difference (Wilcoxon). Core: SPP1, VEGF (hypoxia, macrophages); periphery: PTN, MHC-II (infiltration, talking to normal brain)

The core has more and stronger interactions (hypoxia, macrophage infiltration); only bold pathways in rankNet differ between conditions. One set per patient; write it up only when all four agree

Beyond CellChat: LIANA consensus and other tools

Methods disagree substantially; reviewers often ask for agreement between at least two

Tool	Feature	Output	When
CellChat (this course)	Multi-subunit, co-factors, pathway level richest plots	L–R, pathway, network	Default; when you need the plots
LIANA+	Runs CellPhoneDB, NATMI, Connectome and more, takes a consensus rank	Consensus L–R rank	To cross-validate CellChat
CellPhoneDB	The original; curated human database	L–R + p values	Python pipelines; human data
NicheNet	Asks which ligand best explains target-cell expression change — infers from downstream genes	Ligand activity + target genes	When you have a clear receiver DE signature
Spatial validation	Visium / Xenium: are ligand and receptor actually adjacent	Co-localisation	Before writing 'X drives Y'

08

Trajectory: how states transition

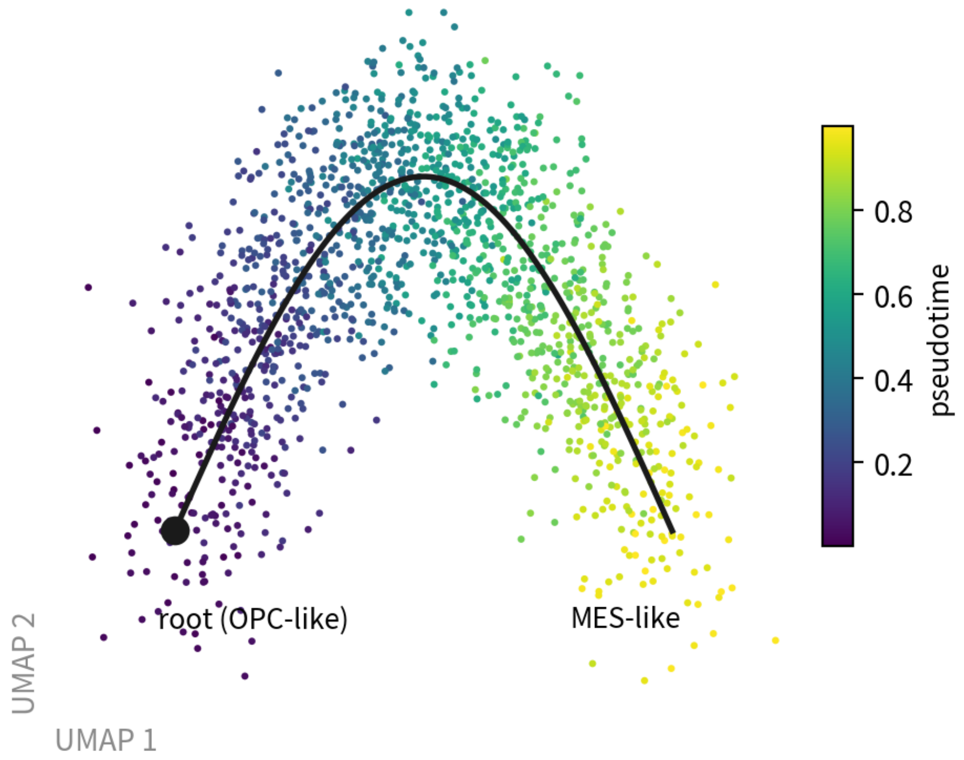
Script 08: Slingshot + tradeSeq mainline; Monocle3 / Monocle2 comparison and hand-picked roots

Trajectory: reading pseudotime and gene dynamics

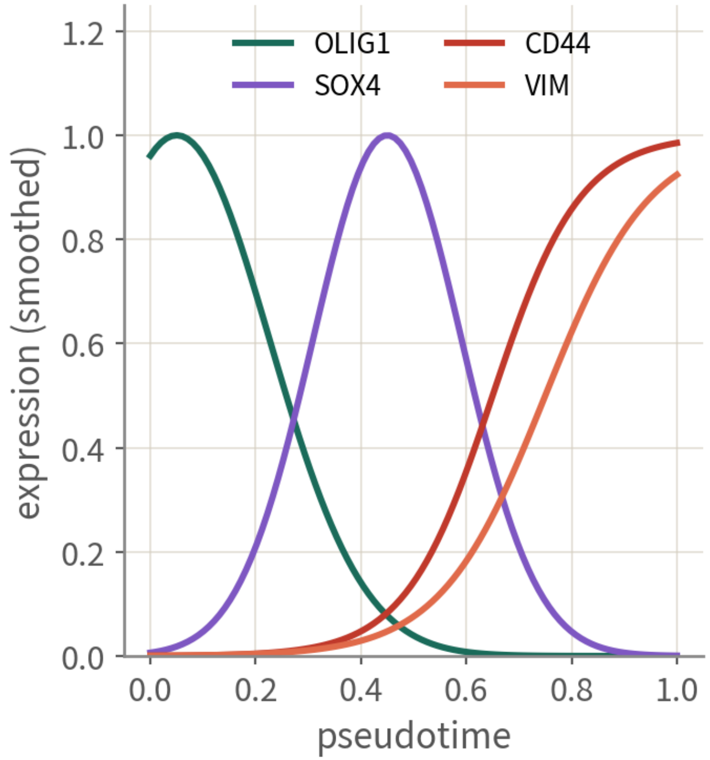
Trajectory analysis: reading pseudotime and gene-dynamics plots

schematic / simulated

Slingshot: principal curve + pseudotime (one patient's malignant cells)



Genes changing along pseudotime (tradeSeq)



Four conditions for a credible trajectory

1. Only within one patient's malignant cells
2. The root has a biological reason (e.g. OPC-like, or RNA velocity points there)
3. Cell cycle does not drive the trajectory (else regress first)
4. Same direction with a new seed and in other patients

Only within one patient's malignant cells; the root needs a reason; the cycle must not drive it; direction consistent across seeds and patients. Fail any of the four and the curve is decoration

Slingshot + tradeSeq: one patient's malignant cells

```
library(slingshot); library(tradeSeq)
mal1 <- subset(gbm4, malignant == "malignant" & patient == "BT_S2")
mal1 <- NormalizeData(mal1) |> FindVariableFeatures() |> ScaleData() |> RunPCA() |>
        FindNeighbors(dims = 1:15) |> FindClusters(resolution = 0.4) |> RunUMAP(dims = 1:15)
sce  <- as.SingleCellExperiment(mal1)
sce  <- slingshot(sce, clusterLabels = "seurat_clusters", reducedDim = "UMAP",
                  start.clus = "2")                        # 起點：OPC 樣分數最高的群
pt   <- slingPseudotime(sce)[, 1]
FeaturePlot(mal1, features = "pt")                         # 先把 pt 存進 mal1$pt

# 沿 pseudotime 變化的基因（負二項 GAM）
counts <- LayerData(mal1, layer = "counts")[VariableFeatures(mal1), ]
gam <- fitGAM(counts = counts, pseudotime = pt, cellWeights = rep(1, ncol(counts)), nknots = 6)
assoc <- associationTest(gam); head(assoc[order(assoc$pvalue), ])
plotSmoothers(gam, counts, gene = "CD44")

# RNA velocity（可選）：需要 loom（velocyto）→ scVelo（Python）；用它決定方向與起點
```

Monocle3 is the other common choice (graph-based, allows branching). Whichever you use, root and direction need independent evidence | Script 08 §1

Monocle3 / Monocle2: two ways to pick the root

```
# Monocle3（最多人用；graph-based）
library(monocle3)
cds <- SeuratWrappers::as.cell_data_set(mal1)   # 沿用 Seurat 的 UMAP 與分群
cds <- cluster_cells(cds) |> learn_graph(use_partition = FALSE)
# (a) 手動互動式：跳出視窗讓你「點」起點——直觀，但不可重現
cds <- order_cells(cds)
# (b) 腳本化：把理由寫成程式碼——OPC 樣分數前 10 顆當 root，可重現
root.cells <- colnames(mal1)[order(mal1$OPC1, decreasing = TRUE)[1:10]]
cds <- order_cells(cds, root_cells = root.cells)
plot_cells(cds, color_cells_by = "pseudotime", label_branch_points = TRUE)

# Monocle2（經典 DDRTree；root_state 一樣是自己選）
cds2 <- reduceDimension(cds2, method = "DDRTree") |> orderCells()
plot_cell_trajectory(cds2, color_by = "State")   # 看圖決定哪個 State 當起點
cds2 <- orderCells(cds2, root_state = 3)         # ← 手動填；腳本版用分數選

cor(mal1$pt, mal1$pt_m3, method = "spearman")    # 跨工具檢查：> 0.8 才可信
```

Pick by hand to explore, script it for the final run — hand-picking is irreproducible. Only conclusions consistent across Slingshot, Monocle3 and Monocle2 go into the paper | Script 08 §2–3

09

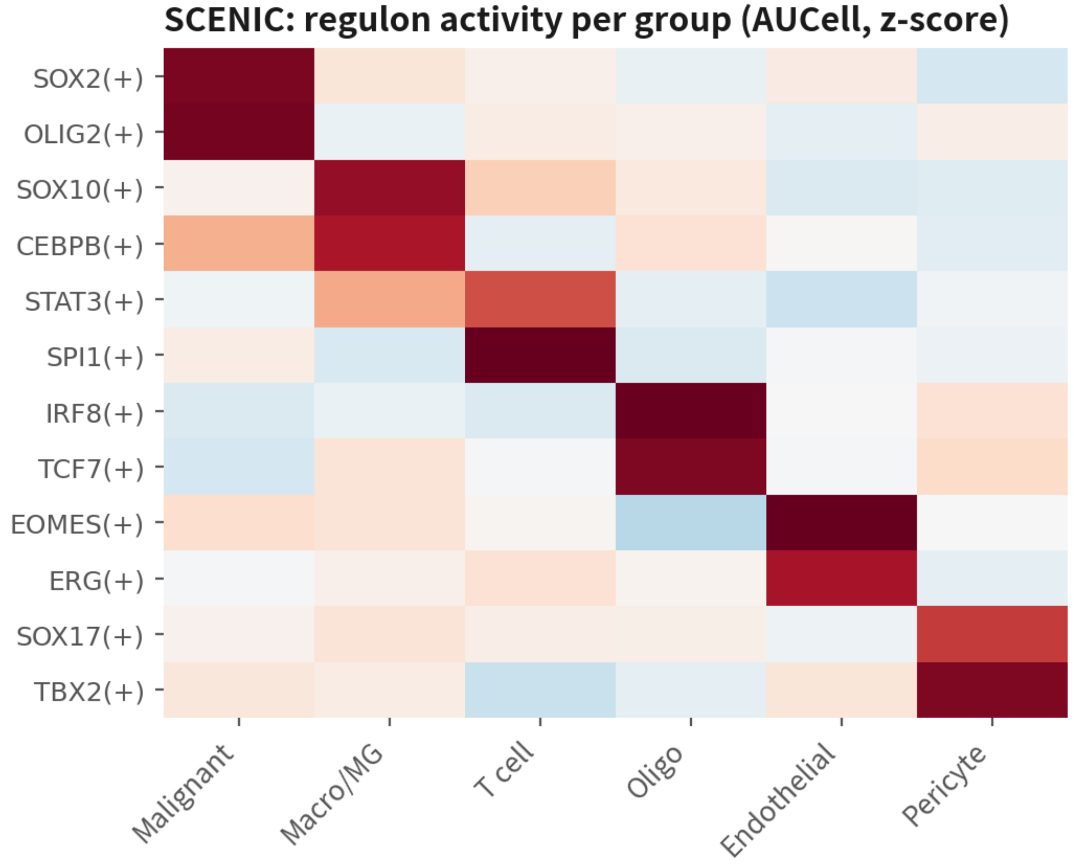
Pathway and TF activity: who drives it

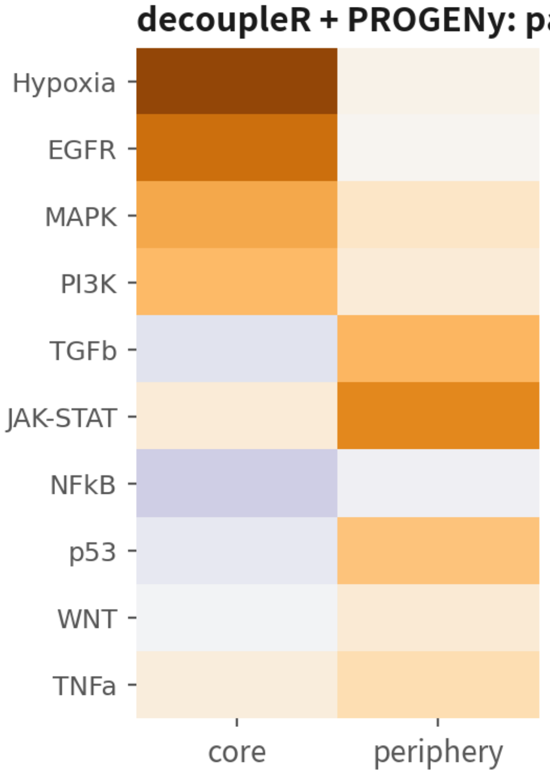
Script 09: decoupleR / PROGENy (R, fast) and SCENIC (Python, slow, optional)

Regulation and activity: who is driving this group

Regulation and activity: from 'which genes changed' to 'who drives it'

schematic / simulated





The difference

SCENIC asks 'which TF drives this group': regulons from co-expression+motifs, then activity per cell
PROGENy/decoupleR asks 'which pathway is on': footprint genes learned from perturbations, not pathway members
Both can be compared between conditions after pseudobulk—the unit is still the patient

SCENIC answers 'which transcription factor', PROGENy / decoupleR answer 'which pathway is active'. Both can be pseudobulked and compared by patient

decoupleR (R) and SCENIC (pySCENIC)

```r
# 1. decoupleR + PROGENy：每顆細胞的 14 條路徑活性
library(decoupleR)
net <- get_progeny(organism = "human", top = 500)
mat <- as.matrix(LayerData(gbm4, layer = "data"))
act <- run_mlm(mat, net, .source = "source", .target = "target", .mor = "weight")
act.w <- tidyr::pivot_wider(act, id_cols = source, names_from = condition,
                            values_from = score) |>
         tibble::column_to_rownames("source") |> as.matrix()
gbm4[["progeny"]] <- CreateAssayObject(act.w)
DefaultAssay(gbm4) <- "progeny"
FeaturePlot(gbm4, features = c("Hypoxia", "JAK-STAT"))
# 條件比較：活性先平均到 病人×部位，再做配對檢定（單位 = 病人）

# 2. SCENIC：R 匯出 loom → pySCENIC 三步 → 讀回 AUCell 矩陣
#   pyscenic grn    counts.loom tfs.txt -o adj.csv      # 共表現（GRNBoost2）
#   pyscenic ctx    adj.csv *.feather --annotations_fname motifs.tbl -o reg.csv
#   pyscenic aucell counts.loom reg.csv -o auc.loom     # regulon 活性
auc <- read.csv("output/09_scenic_auc.csv", row.names = 1)   # regulon × cell
gbm4[["scenic"]] <- CreateAssayObject(as.matrix(auc))
DoHeatmap(gbm4, features = c("SOX2(+)", "OLIG2(+)", "SPI1(+)", "CEBPB(+)"),
          assay = "scenic")
```

SCENIC takes hours per run; the classic GBM result is malignant SOX2 / OLIG2 and TAM SPI1 / CEBPB. The (+) after a regulon means activating | Script 09

10

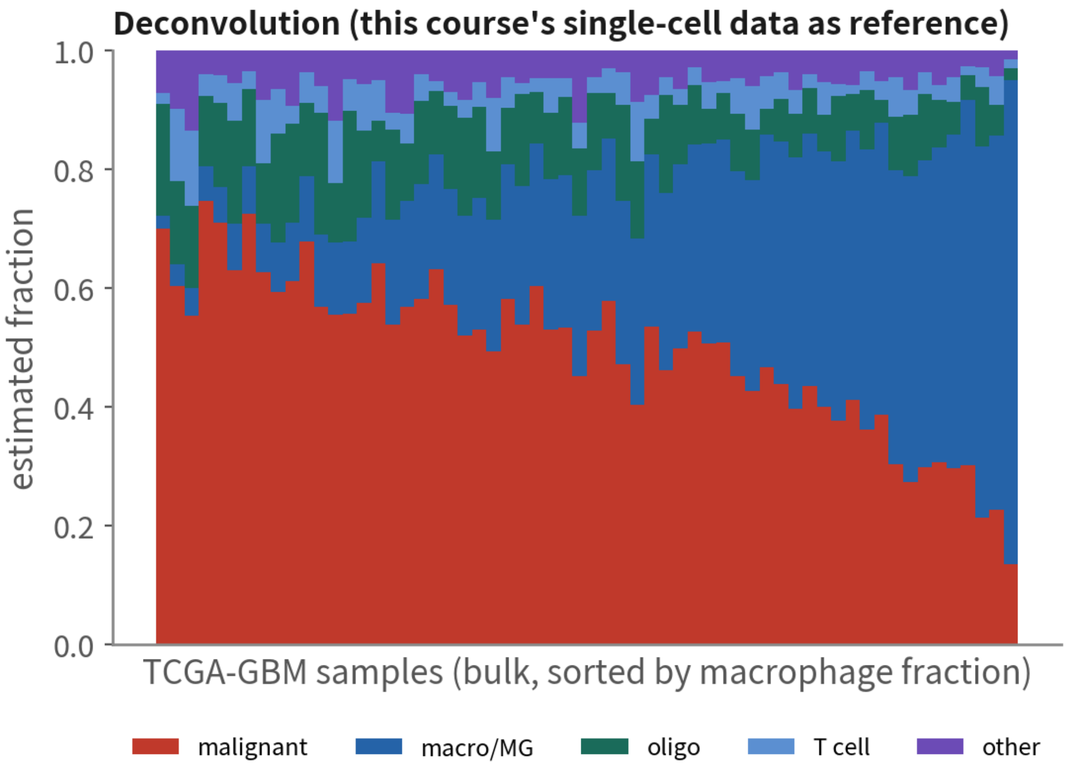
Deconvolution and survival: validate in a large cohort

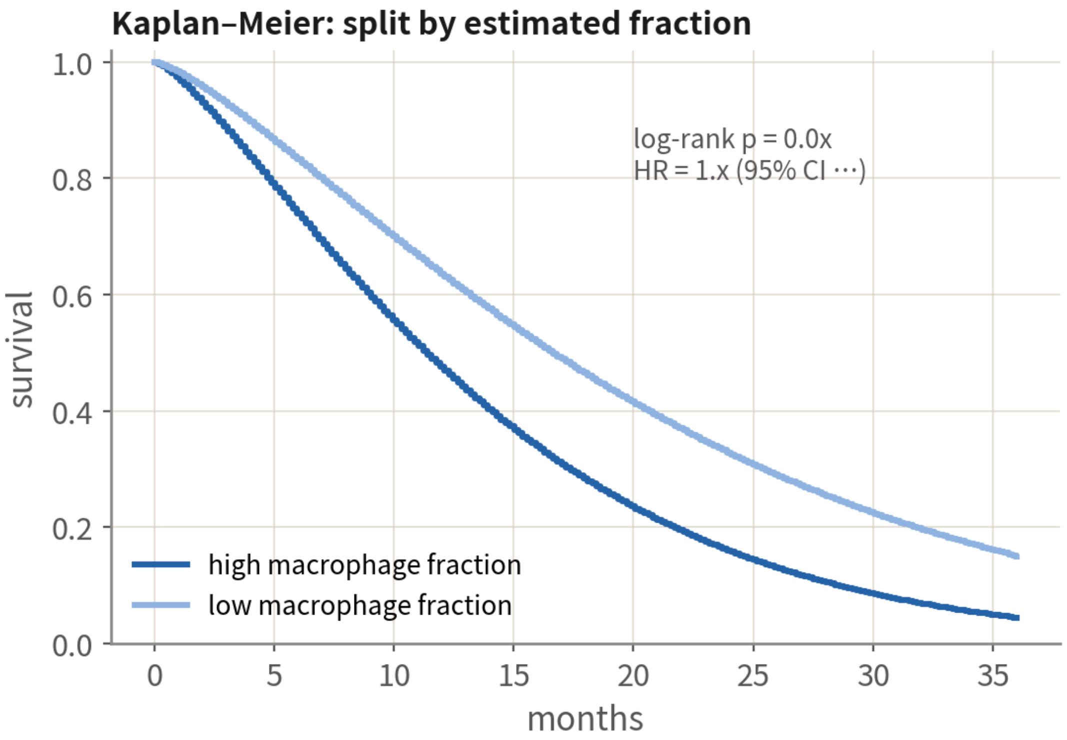
Script 10: MuSiC + TCGA-GBM + KM / Cox — the exit for single-cell hypotheses

Deconvolution: taking n = 4 to n = 500

Deconvolution: carry single-cell type signatures back to hundreds of bulk samples, and buy statistical power

schematic / simulated





The right exit from n=4: generate the hypothesis in single cell (e.g. 'macrophage fraction relates to outcome'), validate in TCGA's n=500. Numbers are illustrative

Use your annotated single-cell data as the reference, estimate type fractions in every TCGA bulk sample, then run survival. The most common — and most convincing — validation route for single-cell studies

MuSiC deconvolution + TCGA survival

```
library(MuSiC); library(TCGAbiolinks); library(survival); library(survminer)
# 1. 參考：本課的單細胞（counts + celltype_l1 + patient）
ref <- as.SingleCellExperiment(JoinLayers(gbm4))
# 2. bulk：TCGA-GBM 的 counts 與臨床
q <- GDCquery("TCGA-GBM", data.category = "Transcriptome Profiling",
              data.type = "Gene Expression Quantification",
              workflow.type = "STAR - Counts")
GDCdownload(q, directory = "C:/GDCdata")          # Windows：路徑上限 260 字元
bulk <- GDCprepare(q, directory = "C:/GDCdata")   # → 用短路徑，否則整批解壓失敗
est <- music_prop(bulk.mtx = assay(bulk, "unstranded"), sc.sce = ref,
                  clusters = "celltype_l1", samples = "patient")
prop <- as.data.frame(est$Est.prop.weighted)      # 樣本 × 型別
# 3. 存活：巨噬比例上下半
clin <- as.data.frame(colData(bulk))[rownames(prop), ]
clin$mac_hi <- prop$Immune > median(prop$Immune)
fit <- survfit(Surv(time, event) ~ mac_hi, data = clin)
ggsurvplot(fit, pval = TRUE, risk.table = TRUE)
coxph(Surv(time, event) ~ mac_hi + age_at_index, data = clin)
```

BayesPrism and CIBERSORTx are alternatives. Adjust the Cox model for age, MGMT and other known factors; fractions are estimates, not truth | Script 10

Downstream toolbox: question → tool → input → minimum evidence

Question	Tool	Input	Minimum evidence before the paper
Who talks to whom	CellChat, LIANA	Same sample, post-CNV labels, log-normalized	Two methods agree, all patients agree expression verified
How states transition	Slingshot, Monocle3 scVelo	One patient's malignant cells	Justified root + consistent across seeds / patients
Who drives it	SCENIC, decoupleR	counts (SCENIC) / data (decoupleR)	Regulon targets also move in the DE
Does it hold in a large cohort	MuSiC, BayesPrism	Single-cell reference + bulk counts	Still significant after adjusting known factors
Where in space	RCTD, cell2location	Single-cell reference + Visium	Ligand and receptor truly adjacent in space

Series Q trap index

Ep.	Trap	One-line fix
Q1	Cells as samples	n is patients; pseudobulk for DE
Q1	Malignancy from markers	Markers give lineage; CNV calls malignancy
Q1	Four malignant islands as four subtypes	Those are four patients; read states within patients
Q2	Copying the 5% mito cut	median + 3×MAD, number in the methods
Q2	Doublet cluster as immune-like tumour	Per-cluster doublet rate after clustering; remove the cluster
Q3	Integration erasing patient-specific malignant biology	Integrated space for normal cells only; malignant in unintegrated
Q3	inferCNV reference contaminated by malignant cells	Reference = immune + oligo; never astrocytes
Q3	Pooling patients for CellChat	Run per sample; report only what all patients agree on
Q3	Trajectories across patients' malignant cells	One patient, after CNV, a justified root

09

Where people get hurt

Three traps that get a manuscript bounced

Trap 1: the wrong inferCNV reference

What it is

Using the 'Astrocyte' cluster as the normal reference because its markers look like normal astrocytes.

Why it happens

Astrocytes are normal brain cells in the textbook.

What it costs you

Most GBM malignant cells are AC-like; with malignant cells in the reference, the chr7/chr10 signal is subtracted away, the heatmap flattens and every call fails.

Trap 2: DE or CNV on integrated values

What it is

Running FindMarkers on core vs periphery right after integration, or feeding integrated values to inferCNV.

Why it happens

The integrated object looks like clean data.

What it costs you

Corrected values are not expression. DE returns to RNA counts and pseudobulk; inferCNV wants raw counts. One check: is the matrix integers?

Trap 3: clinical conclusions from n = 4 pairs

What it is

'Periphery malignant cells express gene X — a therapeutic target for infiltration.'

Why it happens

Paired design, pseudobulk and GSEA all done right, and p looks good.

What it costs you

Right method, insufficient power and generalisability. Four patients give candidates and a consistent direction; conclusions need more patients and independent validation (TCGA deconvolution, spatial data).

Q3 checklist

Q3 checklist: all nine boxes before it goes into a paper

- ✓ unintegrated plot and object saved first
- ✓ integrated space used only to annotate normal types
- ✓ inferCNV reference= immune+oligodendrocytes
- ✓ malignant labels triangulated; unsure= unresolved
- ✓ pseudobulk matrix integer, one sample per column
- ✓ DESeq2 design includes patient (paired)
- ✓ per-sample dots confirm consistent direction
- ✓ GSEA passes the known-answer check
- ✓ data prerequisites for the next road checked

All nine boxes before it goes into a paper

ONE-LINE SUMMARY

**Integrate normal cells only; call malignancy by CNV, per patient;
the unit of DE is the patient; every road after DE needs the right data.**

With the six sentences from Q1 and Q2, that is the skeleton of tumour single-cell analysis.

Check yourself 1

Which group is the safest inferCNV reference?

A The GFAP-expressing astrocyte cluster

B Immune cells and oligodendrocytes

C Whatever sits farthest from malignant cells on the UMAP

Answer B. The reference must be certainly CNV-free. Immune cells and oligodendrocytes are essentially never malignant in GBM; astrocytes may be AC-like malignant cells; UMAP distance is not evidence.

Check yourself 2

After integrating four patients, malignant cells are one blended cluster. This means?

A Integration worked; proceed downstream

B Integration erased patient-specific CNV biology; analyse malignant cells in the unintegrated space

C The four patients share a subtype

Answer B. Malignant patient islands are genomic differences, not batch. Use the integrated space only to annotate normal cells.

Check yourself 3

For core vs periphery malignant DE, what should the DESeq2 design be?

A ~ tissue

B ~ patient + tissue

C ~ tissue + n_cells

Answer B. Each patient contributes both sites — a paired design; absorb patient first, then test site. Cell number is not a covariate; pseudobulk already handled it.

Where to go next

Series A (6 ep.)	Design (A3), reading plots (A5), results to story (A6) — what Q1 had no time for
Series B (18 ep.)	Every 'Deeper: Bx' tag is the index. Tumour-relevant: B13 enrichment, B14 communication, B16 inferCNV, B17 deconvolution
Series C (8 ep.)	C4 PCA, C7 inference (why pseudobulk works), C8 the maths of integration
Practice scripts R/00–08	Each ends with exercises; finish all nine and you have completed a tumour single-cell analysis on your own

END OF SERIES Q

Two hours, and you have the skeleton of scRNA-seq analysis

Practice scripts R/00–08, quizzes in quiz/, transcripts in docx/.

Now take your own tumour data and grow the skeleton into muscle.